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Anomaly (Out-of-Distribution) Detection for Electrical Distribution Networks using Discrete Diffusion Methods

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Abstract

Electrical distribution monitoring faces a difficult anomaly-detection regime: abundant normal measurements, scarce and heterogeneous labelled faults, quasi-periodic structure with benign transients, and non-stationary drift. This report evaluates diffusion-based *one-class* detectors for high-sample-rate magnetic-field time series, using fixed-length window scoring, suitable for streaming deployment.

The core model is *discrete diffusion* over quantised amplitudes, trained as a categorical denoiser on normal-only windows. We compare uniform amplitude bins to CDF/quantile binning and score windows using (i) single-pass reconstruction residuals at a fixed corruption level and (ii) a likelihood proxy, using Monte Carlo denoising-loss.

Empirically, the model reconstructs normal structure well, but fault–normal score separation remains weak. In a 36-run model-selection sweep, the best likelihood-scoring configuration achieves only modest gains over chance (sequence-level AUROC 0.548, PR-AUC 0.336; prevalence 0.279). Reconstruction-based scoring shows similarly limited separability (AUROC 0.515, PR-AUC 0.347). CDF/quantile binning tends to help likelihood-style scores, while uniform binning can help reconstruction PR-AUC. When thresholds are calibrated to respect a false-alarm budget, operating points still imply unrealistically high alert rates. Persistence-based policy filters reduce false alarms but also suppress detections. These findings indicate that score stability/separability (not threshold choice alone) is the primary bottleneck.

Terminology. This report uses *anomaly detection* for the deployed objective and *out-of-distribution (OOD) detection* when connecting to the broader ML literature; *drift* refers to temporal distribution shift in the normal stream.

Preface (submission mapping). This submission is organised in two parts: Chapters 1–2 form the essay (background and motivation), and Chapters 3–11 form the project report (system, experiments, results, and deployment).

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Contents

I Background and motivation **1**

1 Introduction **3**

2 Problem domain and modelling foundations **5**
2.1 Central claim and modelling stance 5
2.1.1 Operational framing: one-class sequential detection 5
2.2 Electrical distribution systems and magnetic field measurements 6
2.2.1 Signal characteristics 6
2.2.2 Fault signatures and temporal evolution 8
2.2.3 Return paths, grounding practice, and the detection gap 8
2.2.4 Noise and non-stationarity 10
2.3 Classical and feature-based approaches to fault detection 10
2.3.1 Rule-based and thresholding methods 10
2.3.2 Feature engineering and classical signal processing 11
2.3.3 Conventional machine learning classifiers 12
2.3.4 Why these limitations matter for high-impedance faults 12
2.3.5 Transition to generative and diffusion-based approaches 12
2.4 Diffusion models for time-series data 13
2.4.1 Generative modelling intuition 13
2.4.2 Signal representation, preprocessing, and invariances 14
2.4.3 Diffusion models at a high level 16
2.5 Discrete diffusion for anomaly detection 17
2.5.1 Motivation: changing the geometry of mismatch 17
2.5.2 Discrete diffusion formulation 18
2.6 Anomaly detection using diffusion-based models 20
2.6.1 Scoring anomalies in generative models 20
2.6.2 Sliding windows for sequential anomaly detection 21
2.6.3 Evaluation criteria under imbalance and drift 21
2.6.4 Related work: diffusion models for time-series modelling and anomaly detection 22
2.7 Operational constraints for anomaly detection systems 23
2.7.1 Latency and temporal context 23
2.7.2 Compute and throughput constraints 24
2.7.3 Robustness to drift and update cadence 24
2.7.4 Interpretability, diagnostics, and trust 24
2.7.5 False positives and operational cost 25
2.7.6 Integration with downstream decision logic 25

II System, experiments, and deployment **27**

3 Problem formulation and system overview **29**
3.1 Problem formulation: one-class sequential anomaly detection 29
3.2 System overview 29

4 Data and preprocessing **31**
4.1 Laboratory dataset format 31
4.1.1 Channel definitions ($C = 16$) 31
4.2 Train/test separation 31
4.3 Windowing 32
4.4 Fault labels and masks 32
4.4.1 Label integrity / labelling validation 33
4.5 Normalisation 33
4.6 Quantisation for discrete diffusion 34
4.6.1 Uniform binning 34
4.6.2 CDF/quantile binning (training-fit) 34

5 Discrete diffusion model for time-series **37**
5.1 Tokenisation of time-series windows 37
5.2 Forward (corruption) process 37
5.2.1 Biased timestep sampling 38
5.3 Denoising model 38
5.4 Training objective and optimisation 38
5.5 Extensions (optional) 38
5.5.1 Contrastive learning with pseudo-anomalies 38

6 Anomaly detection and scoring methods **41**
6.1 Problem setting 41
6.2 Reconstruction-based anomaly score 41
6.3 Likelihood / ELBO-proxy anomaly score 42
6.3.1 Discrete diffusion 42
6.4 Extensions (optional) 42
6.4.1 Gradient-based sensitivity scoring 42
6.4.2 Inpainting-based anomaly amplification 43
6.5 Aggregation views and causal post-processing 43
6.6 Threshold selection and detection latency 43

7 Experimental design **45**
7.1 Configuration-driven experiments 45
7.2 Data settings 45
7.3 Model variants 46
7.4 Training protocol 46
7.5 Evaluation protocol 46
7.5.1 Baselines and comparisons 46
7.5.2 Automated model selection and score-family choice 47
7.5.3 Run orchestration and robustness 48
7.5.4 Compute budget and reproducibility 48
7.5.5 Fidelity-only mode 48
7.6 Evaluation metrics 48
7.6.1 Fidelity diagnostics 48

7.6.2 Anomaly-detection metrics . . . 49
7.6.3 Operating-point protocol . . . 49
7.6.4 Validity controls . . . 49

8 Results and evaluation **51**
8.1 Model selection sweep (lab dataset) . . . 51
8.1.1 Score families, views, and selection criteria . . . 52
8.1.2 Threshold-free results across the grid . . . 52
8.1.3 What worked (and why) . . . 52
8.1.4 Selected configuration . . . 53
8.1.5 Qualitative diagnostics and failure cases . . . 54
8.1.6 Operational and sensitivity ablations (summary) . . . 55
8.1.7 External baseline: supervised fault classifier (prior work) . . . 56
8.2 Benchmark datasets (MSL/SMAP/PSM/SMD) . . . 57

9 Real-time performance and deployment considerations **63**
9.1 Streaming interface and latency budget . . . 63
9.2 What makes inference expensive (and what does not) . . . 63
9.2.1 Single-pass reconstruction at a fixed timestep . . . 63
9.2.2 Likelihood-style scoring as Monte Carlo denoising error . . . 64
9.2.3 Gradient and inpainting compute costs . . . 64
9.3 Batching and hardware considerations . . . 64
9.3.1 Raspberry Pi 5: CPU-first optimisation . . . 64
9.4 Causal post-processing and alerting . . . 65
9.5 Thresholding, calibration, and drift . . . 65
9.6 Summary . . . 65

10 Discussion and limitations **67**
10.1 Implementation pipeline . . . 67
10.2 Reconstruction scores versus likelihood-style scores . . . 67
10.3 Discrete diffusion in this project . . . 68
10.4 Windowing, smoothing, and the detection-latency trade-off . . . 69
10.5 Evaluation limitations and threats to validity . . . 69
10.6 The role of contrastive learning . . . 70
10.7 Summary . . . 70

11 Conclusion and future work **71**
11.1 Summary of contributions . . . 71
11.2 Findings . . . 71
11.3 Future work . . . 72

A Supplementary ablation results **73**
A.1 Operational sweeps from rerun-all (2026-02-05) . . . 73
A.2 Initial-noise sweep: is ρ=0.5 too low? . . . 75

Figures

3.1 End-to-end pipeline: offline preprocessing and diffusion training on presumed-normal windows, online window-level scoring with temporal decision logic, and evaluation using both reconstruction fidelity and detection metrics. . . . 30

4.1 Distribution of fault durations (seconds) by group: LI vs HI and BC vs Non-BC. Durations are computed from contiguous label masks and file timestamps; sample sizes and medians are reported in the figure caption in the main text. 33

5.1 Discrete denoiser architecture: input signals are quantised to tokens, corrupted via forward diffusion, embedded, and processed by a 1D residual U-Net conditioned on timestep t. Skip connections preserve spatial information across encoder-decoder levels. Self-attention layers capture long-range dependencies in the bottleneck and decoder. The output is a categorical distribution over K bins for each position. 40

8.1 Model-selection sweep (lab dataset): sequence-level threshold-free metrics (AUROC vs PR-AUC) for reconstruction and likelihood score families. Each point is one configuration; colours/markers encode quantiser strategy and repeated-run index. The dashed PR-AUC baseline equals the label prevalence. The star marks the selected configuration (best likelihood PR-AUC). . . 52

8.2 Model-selection sweep (lab dataset): mean sequence-level AUROC and PR-AUC as a function of quantisation bins K and token embedding dimension d_{embed}, averaged over quantiser strategy and repeated-run index. 53

8.3 Distributional diagnostic: reconstruction-error distribution (sequence-level mean score) for normal vs fault windows. A strong detector shows clear separation (fault scores shifted higher with limited overlap). 54

8.4 Distributional diagnostic: reconstruction-error distributions for fault windows split by fault type code (top groups by support). This highlights fault classes that are consistently more or less separable under residual scoring. 55

8.5 Qualitative snapshots: true vs reconstructed raw sensor channels (X and Z), absolute error, and per-timestep reconstruction score for representative normal and fault windows with high/low error. Shaded regions indicate labelled fault spans. 56

8.6 Binary confusion matrix for the prior supervised DNN baseline after collapsing all fault categories into a single "fault" class. Counts are derived from the row-normalised confusion matrix and per-class supports in `cfat_DNN_confusion-matrix.docx`. 57

8.7 Prior supervised DNN baseline: fraction of each fault class predicted as `No-Fault` (binary false negatives), annotated with the corresponding miss counts. . . . 58

8.8 Benchmark protocol comparison (aligned-last): Gaussian vs discrete diffusion for reconstruction and likelihood scores, showing raw pointwise F1 vs point-adjusted (PA) F1. High PA F1 with near-zero raw F1 indicates that the detector often triggers somewhere within anomaly segments but does not localise anomalies well at the point level. 59
8.9 Benchmark rerun (2026-02-05, discrete diffusion): point-adjusted F1 (PA) by dataset and score family. 60
8.10 MSL window-length sweep (Gaussian diffusion, aligned-last): AUROC and point-adjusted F1 (PA) as a function of window length L. 61

A.1 Threshold-calibration sweep (rerun-all): FAR/day(normal) versus detected-event fraction for reconstruction and likelihood scores. Points are annotated by quantile q. 74
A.2 Alert-policy sweep (rerun-all): FAR/day(normal) versus detected-event fraction. Lines connect cooldown settings within each $(K, \text{persistence})$ group. . . 74
A.3 Initial-noise sweep: threshold-free detection metrics vs reconstruction corruption fraction ρ. Reconstruction metrics change with ρ; likelihood metrics are largely invariant because they do not depend on the fixed t_{rec}. The dashed PR-AUC baseline equals the label prevalence. 75
A.4 Initial-noise sweep: reconstruction score distributions for normal vs fault windows (sequence-level sequence-max view). Violins are derived from stored score quantiles rather than raw per-window scores. Heavy overlap and upper-bound saturation help explain why separability remains limited even when PR-AUC increases with ρ. 76
A.5 Initial-noise sweep: reconstruction fidelity (MAE) vs ρ for normal and fault windows, and the resulting MAE gap. Negative gap means faults reconstruct as well as (or better than) normal windows under MAE. 76
A.6 Initial-noise sweep: reconstruction PR-AUC split by fault categories (where available in the dataset labels). The dashed baseline equals the overall label prevalence. 77

Part I

Background and motivation

Chapter 1

Introduction

Reliable fault detection in electrical distribution networks is central to safety and service continuity. This report studies anomaly detection from high-sample-rate magnetic-field time series recorded by non-invasive sensors near distribution infrastructure. Low-impedance faults often produce abrupt, obvious changes; high-impedance faults can evolve gradually and with subtle signatures that overlap benign operational transients [1, 4]. That difference motivates learning the distribution of normal behaviour instead of relying on fixed thresholds or a narrow fault catalogue.

In early work with this laboratory dataset, a supervised classification pipeline was attempted, but it encountered the expected deployment barriers: labelled fault windows were limited and heterogeneous, and models trained on staged faults did not generalise cleanly to new operating regimes [9, 1]. This pushed the project away from fault-type classification and toward a one-class framing in which the model learns normal behaviour and flags deviations.

The core research problem is to develop and evaluate diffusion-based generative models for one-class sequential anomaly detection in distribution monitoring. Diffusion models learn to reverse controlled corruption on fixed-length windows, which suits complex multimodal waveforms [44, 22]. In this setting, windows are scored by mismatch with learned normality using likelihood-style surrogates, reconstruction behaviour, or stepwise denoising diagnostics [34, 17, 21]. Compared with adversarial approaches, diffusion training is also more stable and its score components are easier to inspect under drift.

Anomaly detection vs OOD terminology. Throughout this report, *anomaly detection* refers to the deployed objective: fault-related deviations are treated as rare departures from the learned normal distribution. The term *OOD detection* is used when relating this setup to the broader ML literature and when discussing distribution shifts that may occur without labelled faults (e.g., new operating regimes or sensor drift).

The report makes four main contributions:

- **A window-level framing for HIF detection under drift:** treat detection as calibrated scoring over fixed-length windows, rather than pointwise thresholding or supervised classification over a fixed fault list.
- **An evaluation of discrete diffusion in this regime:** test whether quantisation changes mismatch geometry in ways that better fit heavy-tailed, heteroskedastic magnetic-field amplitudes, using normal-only calibration stability plus detection and fidelity metrics.
- **A clearer separation of score families:** distinguish reconstruction-residual scoring from likelihood-style / denoising-loss scoring, and use stepwise score decompositions

as diagnostics.

- **A deployment-oriented evaluation protocol:** make false-alarm budget a first-class objective, calibrate thresholds on normal-only data, and treat max-F1 oracle thresholds as optimistic diagnostics only.

The remainder of the report is organised as follows. Chapter 2 introduces the sensing context, signal characteristics, and modelling foundations, and motivates diffusion-based approaches for non-stationary time-series anomaly detection. Subsequent chapters describe the system formulation and data preprocessing, develop the discrete diffusion model and scoring methods, and present experimental design, results, and deployment considerations.

Chapter 2

Problem domain and modelling foundations

The discussion in this chapter sets the domain context and modelling foundations for diffusion-based anomaly detection. It covers signal characteristics, relevant generative approaches, and deployment constraints.

2.1 Central claim and modelling stance

The core claim is that *window-level distribution learning* is the right abstraction for magnetic-field high-impedance fault detection under drift and label scarcity [14]. Normal behaviour is quasi-periodic, multi-axis, and multi-modal, while target faults often emerge gradually across time. In this regime, a detector is better treated as a calibrated *scoring model* over windows than as a pointwise classifier or fixed threshold rule. This supports downstream alerting tied to an explicit false-alarm budget.

The report treats the following as a testable hypothesis: *discrete diffusion can provide stable and interpretable anomaly scores in this domain under fixed training budget and model capacity*. The motivation is practical. Gaussian amplitude modelling induces squared-error geometry, and its likelihood surrogates can be dominated by nuisance variance, heavy tails, and heteroskedasticity [34, 17]. Quantisation changes this geometry so mismatches become bin-level deviations with clearer operational meaning and potentially more stable normal-only calibration. Under this stance, pointwise rules, raw reconstruction MSE, and label-dependent thresholding are not sufficient as deployment strategy on their own.

2.1.1 Operational framing: one-class sequential detection

The deployment objective in this report is one-class sequential anomaly detection. The model is trained on windows from presumed-normal operation and then asked to produce a scalar anomaly score for each incoming window in a stream. In machine learning terms, this overlaps with OOD detection, but the operational target is anomaly flagging under a fixed alert budget rather than taxonomy-level fault classification.

Let $x(t) \in \mathbb{R}^C$ denote a multichannel magnetic-field stream and let fixed-length windows be extracted with length L and stride S:

$$x^{(n)} = x_{:,\, nS:nS+L-1} \in \mathbb{R}^{C \times L}.$$

At inference time, the detector maps each window to a score $s(x^{(n)})$, which is then converted into alerts by thresholding and persistence logic.

This framing has several implications that matter for model design and evaluation:

- **Normality is multi-modal.** Benign behaviour includes switching transients, load changes, and regime-dependent waveform variation, so normality should be modelled as a broad distribution rather than a narrow waveform template.
- **Normality is site dependent.** Geometry, grounding, and sensor placement change the observed baseline. Transfer across sites should be treated as a recalibration problem, not assumed by default.
- **Anomaly is operational.** Alerts indicate windows that are implausible under learned normal behaviour and worth investigation; they are not proof of a specific physical fault mechanism.
- **Windowing sets the latency-context trade-off.** Larger windows improve sensitivity to gradual signatures but delay earliest detection. Short windows respond quickly but increase score variance and can miss slowly evolving faults.
- **Drift is expected.** Score distributions can move over time due to sensor and network changes, so thresholding must support normal-only recalibration under distribution shift [14, 50].

This operational framing is used throughout the project chapters as a constraint on both scoring design and evaluation protocol.

2.2 Electrical distribution systems and magnetic field measurements

The next section introduces the electrical distribution environment and the non-invasive magnetic sensing setup used in this report.

2.2.1 Signal characteristics

What is being measured

The magnetic field generated by current flowing through electrical distribution lines provides a non-invasive proxy for system behaviour. In distribution monitoring settings, electromagnetic sensors can be used to measure the local magnetic field in the vicinity of power poles, allowing changes in current flow to be inferred without direct electrical contact. This sensing approach is particularly attractive for real-world deployment, as it avoids intrusive modifications to existing infrastructure and enables sensors to be mounted externally, for example near the base of a pole.

In laboratory studies, this sensing approach is grounded in prior work that experimentally characterises low-voltage (400 V) high-impedance faults using current measurements alongside non-contact giant magneto-resistive (GMR) magnetic sensors placed at a distance from overhead lines [42, 41].

Unlike voltage or current measurements obtained through direct electrical connections, magnetic field measurements capture the aggregate effect of current flow without requiring access to the conductor itself. This makes them well suited to monitoring legacy infrastructure where physical access may be limited or where safety and regulatory constraints prohibit invasive instrumentation. However, this indirectness also means that the measured signal reflects a superposition of effects, including the geometry of the conductor, the return paths of current, and the spatial relationship between the sensor and the electrical system.

As a result, magnetic field measurements encode system behaviour in a distributed and spatially dependent manner. Changes in current magnitude, phase, or return path alter the surrounding magnetic field, but these changes are not mapped one-to-one onto a single observable quantity. Instead, they manifest as structured variations across multiple axes and over time, requiring models that can learn these relationships from data rather than relying on explicit physical inversion.

Axes and multi-channel structure

Magnetic-field sensing is inherently vector-valued: the field has both magnitude and direction, and what is observed depends on the sensor axes and the line geometry. In the 400 V laboratory facility described by [41], the subsequent fault characterisation study uses a 3-axis GMR sensor head and describes the measurement axes in relation to the overhead line geometry (with current flow along the span direction and other axes aligned to physical features such as the cross-arm and conductor sag) [42].

In practice, datasets may be provided as multiple sensor-derived channels per time step, including raw axis measurements and derived quantities (e.g., envelope or time–frequency features). Treating the signal as a multivariate time-series $x(t) \in \mathbb{R}^C$ and preserving channel separation allows models to learn axis-dependent and geometry-dependent structure from data while avoiding assumptions that may not hold uniformly across deployments.

Sampling and time-series nature

The magnetic field signals are sampled at a fixed high rate in the laboratory setup [42], producing high-resolution time-series data. This sampling regime is sufficient to capture the fundamental frequency of the alternating current as well as higher-order harmonics and transient effects introduced by load changes and fault behaviour. At this resolution, successive samples are strongly temporally correlated, reflecting both the periodic nature of the underlying electrical signal and the inertia of the physical system.

Rather than consisting of independent observations, the data form structured sequences in which local context is critical. Short-term correlations capture oscillatory behaviour and harmonics, while longer-term dependencies reflect gradual changes in operating conditions or fault evolution. Any modelling approach that treats samples independently risks discarding this structure and mischaracterising normal variability as anomalous.

From an anomaly detection perspective, this time-series structure implies that deviations of interest may be distributed across many samples rather than appearing as isolated spikes. Effective detection therefore requires models that can integrate information over time and assess sequences as coherent units, rather than relying solely on instantaneous measurements.

Why this signal is non-trivial

At a superficial level, the magnetic field measurements produced by an electrical distribution system may appear highly regular. The alternating current driving the network induces a strong quasi-periodic component at the fundamental line frequency, along with a predictable set of harmonics. If the system were perfectly stationary, anomaly detection could plausibly be reduced to identifying deviations from a fixed periodic template.

In real operation, the signal departs from this idealised picture. Load changes occur continuously and introduce transient fluctuations in current draw, which appear as amplitude modulation, phase shifts, and short-lived spectral changes in the magnetic field [42, 41]. These transients are normal, not exceptional.

The resulting distribution is structured but not stationary: periodic behaviour, transients, and longer-term drift are all present. Fault signatures are superimposed on this background and can overlap benign behaviour in both time and frequency.

This overlap makes the signal particularly challenging for threshold-based or rule-based detection methods. Simple statistics such as instantaneous amplitude or energy may fluctuate substantially during normal operation, while high-impedance faults may produce only subtle deviations that are distributed over long time horizons. Detecting such faults therefore requires models that can integrate information over time and reason about deviations from a learned distribution rather than from a fixed baseline.

From a modelling perspective, this motivates approaches that explicitly represent the full distribution of normal behaviour rather than attempting to define a small set of hand-crafted features or rules. Generative models are well suited to this setting, as they aim to capture the variability inherent in the data itself. Anomalies can then be identified as samples that are unlikely under the learned distribution or that are reconstructed poorly by the model, without requiring explicit specification of fault signatures.

2.2.2 Fault signatures and temporal evolution

Fault events in electrical distribution systems exhibit a wide range of temporal characteristics depending on their underlying physical mechanisms. Low-impedance faults, such as direct short circuits, typically produce abrupt and large-magnitude changes in current. These events manifest as immediate and pronounced deviations in the measured magnetic field and are often straightforward to detect using rule-based or threshold-based methods.

High-impedance faults present a more challenging detection problem. Rather than producing a single extreme deviation, these faults often evolve gradually over extended periods, sometimes exceeding a minute in staged-fault studies [9, 1]. The fault progression is commonly described in qualitative stages such as sizzling, crackling, and arcing, reflecting intermittent and unstable conduction paths [1, 4]. Throughout this progression, changes in the magnetic field are subtle, temporally distributed, and may vary in both amplitude and spectral content.

This staged evolution and the feasibility of observing fault-related structure using non-contact magnetic sensing in a 400 V physical simulation facility are supported by the laboratory characterisation in [42].

In this magnetic-sensing setting, a recurring characteristic is the gradual emergence of asymmetry across sensing axes [42, 41]. As current increasingly returns through unintended paths, the spatial distribution of the field changes in ways that are not captured by instantaneous magnitude alone. These changes may be weak at any single time step but become more pronounced when viewed over longer temporal windows.

From a modelling perspective, this temporal evolution implies that fault detection cannot rely solely on detecting instantaneous outliers. Instead, it requires assessing how sequences evolve over time and identifying deviations in structure, consistency, or symmetry relative to normal operation. Models that explicitly represent temporal dependencies are therefore essential for capturing the gradual onset and progression of high-impedance faults.

2.2.3 Return paths, grounding practice, and the detection gap

A central reason high-impedance faults are difficult to detect is that they break a simplifying assumption embedded in many traditional protection schemes: that faults produce sufficiently large fault current to be reliably detected by magnitude-based elements. In over-

head distribution systems, many protection mechanisms are tuned around low-impedance faults, where a conductor-to-ground or phase-to-phase short produces a sharp increase in current and a corresponding change in measured electrical quantities. These events tend to be "loud" in both time and amplitude.

High-impedance faults are "quiet" by comparison. When a conductor contacts a high-resistance surface such as dry soil, asphalt, vegetation, or a poorly conducting interface, the resulting current may be too low to trigger conventional overcurrent protection. Industrial relay and protection literature repeatedly highlights that grounded distribution systems can experience downed-conductor or arcing faults whose current magnitude is insufficient for traditional ground overcurrent elements, motivating detection approaches that rely on characteristics beyond magnitude alone [1, 4].

The physical mechanism is also qualitatively different. Rather than forming a stable, low-resistance connection, high-impedance faults often involve intermittent conduction paths and arcing activity. The contact interface can heat, carbonise, and reconfigure over time, producing non-linear and time-varying impedance. This creates signatures that may be distributed across time and frequency, and which can change as the fault evolves. In laboratory and field characterisations, the evolution is often described in stages such as "sizzling" and "arcing", reflecting changes in conduction regime and the emergence of additional frequency content as the fault progresses.

From the perspective of non-invasive magnetic-field sensing, this has two implications.

First, the absolute field magnitude may not exhibit a clean, abrupt step that separates normal from fault. Early-stage high-impedance faults may introduce only subtle deviations that are comparable in scale to benign load-driven transients. Any approach that relies on single-threshold excursions is therefore inherently exposed to a precision–recall trap: lower the threshold and drown in false alarms; raise it and miss early-stage faults.

Second, changes in current return geometry become more informative than changes in magnitude alone. When current returns through unintended paths (soil, vegetation, pole structure), the spatial distribution of current around the pole changes. Magnetic-field sensors do not observe "current" directly; they observe a field that is shaped by both current magnitude and geometry. This creates an opportunity: even if total current change is modest, a redistribution of return paths can manifest as axis-dependent changes, altered cross-axis relationships, or systematic asymmetry that is less likely to be produced by benign load switching.

The price of this opportunity is that geometry also introduces nuisance variability. Sensor placement differences, pole layout differences, and slow drift in mounting orientation can all change the mapping between current geometry and measured axis projections. This is one reason a fully hand-engineered "symmetry rule" is brittle: the definition of symmetry is sensor- and geometry-dependent. A learned distribution over windows can absorb stable, site-specific asymmetries as normal while still flagging changes in asymmetry that depart from the learned pattern.

This observation connects cleanly to the modelling choice later in the chapter: rather than attempting to define universal fault features for all deployments, the objective is to learn the normal distribution for a given sensing configuration and then quantify deviations from that learned normality. Generative models provide the machinery to do this, and diffusion models provide a particularly stable way to learn such distributions over high-dimensional windows.

2.2.4 Noise and non-stationarity

In addition to structured variability arising from system operation and fault behaviour, the magnetic field signals are affected by multiple sources of noise and non-stationarity. Short-term noise arises from environmental electromagnetic interference, sensor electronics, and ambient electrical activity unrelated to the monitored conductor. These effects introduce fluctuations that may not correspond to meaningful changes in system behaviour.

Beyond additive noise, the signal exhibits non-stationary behaviour over longer timescales. Changes in operating conditions, such as seasonal load variation or network reconfiguration, alter the statistical properties of the signal even in the absence of faults. Sensor drift and gradual changes in sensor placement or orientation can further modify the observed distribution over time.

These effects complicate anomaly detection, as deviations in the signal may arise from benign measurement artefacts or slow distributional drift rather than genuine faults. Methods that assume fixed mean, variance, or spectral characteristics are therefore prone to false positives or degraded performance over time.

In practice, the detector must separate transient noise, gradual non-stationarity, and fault-related structure. This motivates sequence-level distribution modelling rather than static thresholds with fixed stationarity assumptions.

2.3 Classical and feature-based approaches to fault detection

Before moving to diffusion-based modelling, it helps to place this approach against established fault-detection practice. Much of the existing work relies on thresholding, handcrafted signal features, and conventional classifiers. These methods are still widely used because they are cheap and easy to interpret. In this sensing regime, though, their limits show up quickly, especially for high-impedance faults.

2.3.1 Rule-based and thresholding methods

The most direct approach to fault detection is to define thresholds on signal magnitude or energy [1, 4]. For low-impedance faults, which typically induce abrupt and large changes in current, thresholding schemes can be effective: spikes in amplitude, sudden increases in RMS current, or step-like deviations in derived statistics can be detected with low latency and minimal modelling. Similar approaches can be extended with simple persistence criteria (e.g., requiring the threshold to be exceeded for a minimum duration) to reduce sensitivity to transient noise.

In magnetic-field measurements, however, thresholding is complicated by the fact that signal amplitude variations occur during normal operation. Load switching, inrush currents, and benign transients can produce excursions comparable in magnitude to early-stage fault behaviour, particularly for high-impedance faults whose signatures may remain subtle for extended periods. As a result, thresholds that are tight enough to detect gradual faults tend to generate false positives during normal operational variability, while thresholds tuned to reduce false alarms often become insensitive to the very faults of interest.

Moreover, in non-stationary environments, a single fixed threshold is rarely appropriate across time. Seasonal load patterns, network reconfiguration, sensor drift, and variations in mounting geometry can change baseline statistics. To remain usable, thresholding approaches often require either manual retuning or adaptive threshold logic, both of which introduce operational overhead and can still fail under unexpected distribution shifts.

Thresholding is therefore most fragile for HIF detection when benign transients overlap early-fault magnitudes and slow drift forces continual retuning to maintain an acceptable false-alarm rate.

2.3.2 Feature engineering and classical signal processing

A common refinement is to replace raw waveform thresholding with hand-engineered features designed to isolate fault-relevant structure. In electrical fault detection, widely used feature families include [1, 15]:

- time-domain statistics such as RMS, peak-to-peak amplitude, variance, kurtosis, and crest factor;
- harmonic and spectral measures such as total harmonic distortion (THD), harmonic ratios, band-limited energy, and spectral centroid;
- time–frequency features derived from short-time Fourier transforms (STFT) or wavelet decompositions, which attempt to capture transient bursts or non-stationary frequency content;
- symmetry or imbalance features, which measure differences between sensing channels (e.g., axis-wise energy ratios or cross-channel correlation), reflecting the expectation that faults disrupt normal balance.

These methods can be effective when fault signatures are strongly associated with specific, stable changes in particular descriptors. For example, arcing behaviour may introduce broadband energy and increased harmonic content, which can be captured by spectral features. Time–frequency decompositions can also improve detection of brief transient bursts that are difficult to observe in aggregate statistics.

However, feature engineering introduces an implicit assumption: that the fault-relevant structure can be captured by a relatively small set of descriptors that remain robust across operating conditions. In the magnetic-field setting, this assumption is strained for several reasons. First, benign transients can produce time–frequency patterns that resemble fault-related behaviour, particularly under varying load compositions. Second, high-impedance faults evolve through stages, meaning that any single feature set may be sensitive to one phase (e.g., arcing) but weak during earlier phases (e.g., sizzling), when detection is operationally most valuable. Third, axis-dependent behaviour is influenced by sensor orientation and geometry, which can blur the interpretation of hand-designed symmetry features unless sensor placement is tightly controlled.

In addition, engineered features can inadvertently discard information needed to distinguish normal variability from faults. Compression into summary statistics often removes fine-grained temporal structure, such as the consistency of oscillatory behaviour, the evolution of subtle asymmetries, or phase relationships that only become apparent when sequences are analysed as coherent units. This is particularly relevant when anomalies manifest as structural deviations spread over time rather than as instantaneous outliers.

Feature-based pipelines are therefore most fragile when assumed fault descriptors are not stable across operating regimes and sensing geometries, and when anomalies appear as distributed structural deviations rather than large changes in a few summary statistics.

2.3.3 Conventional machine learning classifiers

Feature-based pipelines are often paired with standard machine learning models such as logistic regression, support vector machines, random forests, gradient boosted trees, or shallow neural networks. Under supervised settings with sufficient labelled fault data, these models can learn decision boundaries that separate normal from fault regimes, and they can incorporate large feature sets without requiring explicit threshold selection. More recent time-series classifiers, including deep convolutional baselines (e.g., InceptionTime) and transformer-style architectures, can further improve supervised accuracy when labels are abundant [25, 47, 49].

In practice, however, supervised classification for distribution faults frequently encounters data limitations. Fault events—especially high-impedance faults—are comparatively rare, difficult to capture in diverse real-world conditions, and expensive to label reliably [9]. This scarcity leads to class imbalance and weak coverage of the full space of fault manifestations. Models trained on limited fault examples often learn brittle correlations and fail to generalise to unseen fault conditions or new operating regimes.

Even when models perform well on in-distribution test sets, their behaviour under distribution shift is a major concern for deployment. Changes in sensor placement, local electromagnetic environment, or load composition can push feature distributions outside the range observed during training. Discriminative classifiers can then become overconfident or degrade silently, producing either false positives (if they interpret normal drift as fault-like) or false negatives (if faults appear as unfamiliar but not class-separable patterns under the learned decision boundary).

Another limitation is that many classifiers provide scores that are not inherently calibrated as probabilities of fault under changing conditions. Without careful calibration and continuous validation, decision thresholds become fragile. This is operationally costly in infrastructure monitoring, where excessive false alarms reduce trust and missed detections carry safety risks.

Supervised classifiers are therefore strongest when labels are broad and stable; in the scarce, heterogeneous, drifting regime considered here, they tend to overfit a narrow fault catalogue or silently lose calibration as feature distributions shift.

2.3.4 Why these limitations matter for high-impedance faults

The difficulties above become most acute for high-impedance faults. Their signatures are (i) subtle, (ii) temporally distributed, (iii) stage-dependent, and (iv) embedded in highly variable normal operation. Classical methods often excel when anomalies are large, abrupt, and separable by simple descriptors. High-impedance faults instead require modelling approaches that can represent *normal variability in context*, detect deviations in temporal structure, and remain robust under gradual distributional drift.

For this reason, the approach shifts from hand-crafted rules to direct modelling of the normal distribution. If the model captures quasi-periodicity, benign transients, axis-wise structure, and slow non-stationarity, anomalies can be defined operationally as windows that are unlikely under that distribution or cannot be reconstructed faithfully when projected back toward it.

2.3.5 Transition to generative and diffusion-based approaches

Generative modelling offers a different framing: rather than listing fault signatures or engineering features to isolate them, the model learns a distribution over normal sequences and flags deviations from that distribution. This is especially useful when fault labels are scarce

or incomplete, and when the practical requirement is to detect *unexpected* behaviour rather than classify a fixed catalogue of known faults.

Diffusion models in particular offer an expressive and stable framework for modelling complex sequence distributions. By learning to reverse a controlled corruption process applied to entire windows, diffusion-based models can capture rich temporal dependencies without the restrictive assumptions of classical pipelines. The following sections build on this motivation, introducing diffusion modelling foundations for time-series data and discussing how likelihood- or reconstruction-derived quantities can be used to score anomalies in magnetic-field measurements from electrical distribution networks.

2.4 Diffusion models for time-series data

Magnetic-field signals in this domain make anomaly detection difficult in a specific way: normal operation already includes quasi-periodicity, frequent benign transients, and slow drift. High-impedance faults then emerge gradually inside that moving background. Models that rely on fixed baselines or pointwise thresholds miss too much structure. A distribution-learning approach is a better fit because it models the normal sequence itself and treats departures from that model as anomalies.

2.4.1 Generative modelling intuition

Here, the detector needs to learn *a distribution over windows* that already contains quasi-periodicity, benign transients, and slow drift. The choice of generative family matters because each one encodes different assumptions about what should count as a meaningful deviation.

Autoregressive (AR) models treat one-step (or short-horizon) predictability as a proxy for normality [36]. For high-sample-rate magnetic-field windows, that is often the wrong bias. Error can be dominated by phase shifts, benign switching bursts, and local non-stationarity, while early-stage HIFs can appear as *structural* changes (axis asymmetry, harmonic redistribution, intermittency) rather than large one-step surprises. In practice, AR scores are often noisy and hard to calibrate under drift [14, 50].

Latent-variable models such as VAEs assume normal behaviour sits on a relatively smooth latent manifold [26]. That can help as a compression prior, but it also creates two common failure modes for fault detection: (i) over-smoothing rare-but-normal transients, which raises false alarms, and (ii) reconstructing anomalous windows too well when capacity is high, which reduces separability. VAE-style reconstruction scores can work, but they are brittle unless latent geometry is aligned with operational fault semantics [34, 33].

Normalising flows provide exact densities under continuous parameterisations, but that advantage can become a liability here. Magnetic-field amplitudes are usually heavy-tailed and heteroskedastic across both AC cycle phase and operating regime. In that setting, likelihood can be driven more by nuisance variance than by fault-relevant structure. Flow scores can then move with calibration drift even when waveform structure is still operationally normal.

GANs prioritise sample realism, not calibrated density [2]. For anomaly detection, that usually means adding a separate scoring mechanism (discriminator score, reconstruction path, or feature distance), which brings calibration problems back through another route. For this project, that makes GANs a less direct fit.

Diffusion models are a better match here because they frame distribution learning as controlled *corrupt-and-repair* over full windows [44, 43, 22]. That aligns with the goal of this

report: not just realistic generation, but a scoring signal that reflects how much a window must be repaired to look normal.

2.4.2 Signal representation, preprocessing, and invariances

Practical anomaly detection systems sit at the boundary between raw measurement physics and statistical modelling. Many failures attributed to "model weakness" are in fact failures of representation: the model is asked to learn invariances that could have been made explicit, or conversely it is deprived of structure that matters for distinguishing faults from benign variation. For diffusion-based modelling of magnetic-field time-series, representation choices are particularly consequential because the model learns the distribution of windows rather than individual samples.

Windowing and context length

Because the signal is sampled at a fixed high rate, even short durations contain a large number of samples [41, 42]. Fixed-length windows provide a pragmatic interface between streaming sensor data and sequence models. The window length determines which temporal phenomena the model can represent: very short windows capture oscillatory structure and local transients but may miss gradual fault evolution; very long windows provide richer context but increase computational cost and may mix distinct operating regimes within a single sample.

In the distribution setting, two timescales are especially relevant. The first is the AC cycle and its harmonics, which define the dominant periodic structure. The second is the timescale of operational variation, including load switching transients and the gradual onset of high-impedance faults. Windows must be long enough to represent multiple cycles (to stabilise estimates of harmonic structure and axis symmetry) while not so long that they blur across changes in operating regime. In practice, windowing therefore becomes a modelling choice that trades detection latency against sensitivity to slow-developing anomalies.

Overlapping windows can reduce boundary artefacts and improve temporal localisation, but they introduce correlated anomaly scores and increase inference load. Non-overlapping windows reduce compute but risk missing short-lived events that occur near boundaries. These trade-offs interact directly with operational constraints and are discussed further in the deployment section.

Cycle alignment and phase variation

A recurring nuisance in modelling quasi-periodic electrical signals is phase variability: two windows can be identical up to a small temporal shift yet appear different under pointwise error metrics [42]. Phase misalignment can inflate reconstruction errors even when harmonic structure is preserved, and can cause learned distributions to broaden unnecessarily.

There are several ways to handle this. One is to accept phase as part of the normal distribution and rely on the model's capacity to represent it. Another is to enforce an approximate phase convention during preprocessing, for example by aligning windows to a detected zero-crossing or by using a phase reference derived from the dominant frequency component. A third is to use representations less sensitive to phase (e.g., magnitude spectra) for auxiliary evaluation, while keeping time-domain windows for the primary model. In anomaly detection, the goal is not perfect phase normalisation but avoiding pathological sensitivity to harmless shifts that would generate false alarms.

Normalisation and axis coupling

Normalisation choices determine what the model treats as meaningful variation. Global standardisation (using dataset-wide mean and variance) is simple but can be brittle under distribution shift [14, 50]. Per-window normalisation reduces amplitude variability but may inadvertently remove amplitude changes that are fault-relevant. Per-channel normalisation can equalise X and Z scales but may obscure axis asymmetry if applied incorrectly.

Because axis asymmetry is a plausible fault signature, it is important that preprocessing does not enforce artificial symmetry. A practical compromise is to normalise each channel using statistics computed on normal training data (so the model sees stable scaling), while preserving cross-channel relationships within each window. Where sensor mounting variability induces slow changes in gain or orientation, it may be preferable to include simple calibration factors or to treat drift as an explicit phenomenon rather than erasing it through aggressive normalisation.

Filtering and bandwidth considerations

Although high-rate sampling captures harmonics and transients, not all frequency content is equally informative [42, 1]. Environmental electromagnetic interference, sensor electronics noise, and unrelated ambient activity can introduce high-frequency components that behave like noise from the perspective of fault detection. Light-touch filtering can improve signal-to-noise ratio, but overly aggressive filtering risks removing fault-relevant broadband signatures, especially in later-stage arcing behaviour.

A pragmatic framing is to treat filtering as a way to remove known artefacts (e.g., mains hum leakage patterns or obvious sensor noise bands) rather than as an attempt to engineer fault features. Where filtering is used, it should be justified in terms of preserving the dominant signal structure and reducing nuisance variability rather than "improving separability".

Quantisation for discrete diffusion

Discrete diffusion models require a discrete state space. For real-valued sensor signals, this implies a quantisation step that maps continuous amplitudes to a finite set of bins. Here, quantisation is a modelling choice: the number of bins controls representation fidelity, denoising difficulty, and how interpretable likelihood-style scores are [18, 51].

Coarse quantisation reduces model complexity and can improve robustness to small measurement noise, but it may erase subtle early-stage fault signatures that manifest as small deviations. Fine quantisation preserves detail but increases the size of the categorical distribution and can make learning and inference more expensive. The choice therefore acts as an implicit bias about what differences matter. In anomaly detection, this bias should be aligned with the deployment objective: if early detection of subtle deviations is critical, quantisation must be sufficiently fine to preserve them.

A related design choice is whether quantisation is uniform in amplitude or adapted to the empirical distribution (e.g., using quantile bins). Uniform bins preserve linear interpretability of amplitude differences; quantile bins allocate resolution where the signal spends most of its time. Quantile binning can improve modelling efficiency but complicates interpretation because a one-bin change does not correspond to a fixed physical amplitude difference.

Handling multi-axis structure in the representation

Finally, the representation must encode both axes in a way that preserves their relationship. Treating X and Z as independent channels is the simplest approach and allows the model to learn both shared periodic structure and asymmetric deviations. Alternative representations include adding derived channels (e.g., magnitude and phase proxies, or X-Z and X+Z combinations) to make symmetry and imbalance more explicit. These derived channels can help the model focus capacity on physically meaningful patterns, but they also risk imposing assumptions that may not hold under all geometries.

Axis-separated representations preserve directional information, while evaluation can explicitly consider both axis-wise and cross-axis behaviour. This provides a flexible foundation: models can learn normal symmetry implicitly, and anomaly scoring can be designed to emphasise deviations in symmetry where appropriate.

2.4.3 Diffusion models at a high level

For anomaly detection on magnetic-field windows, diffusion is most useful as a *scale-controlled projection operator*. Corruption removes structure in a known way; denoising then tests what can be recovered under learned normality. The practical question is not whether the model can sample, but whether it assigns *stable mismatch* to windows whose quasi-periodic and cross-axis structure departs from normal under drift.

As a result, diffusion design choices should be evaluated primarily through their effect on scoring: how denoising errors decompose across noise levels, whether the score is dominated by nuisance variance, and whether thresholds calibrated on normal data remain meaningful as operating conditions drift.

Architectures and conditioning

Architectures matter here because they implement the inductive biases that decide which deviations are treated as nuisance and which are treated as mismatch.

Convolutional U-Net style denoisers impose a strong *locality + multiscale* bias: early layers model cycle-scale waveform shape and harmonics, while deeper layers capture slower changes in envelope, symmetry, and regime. Skip connections favour preserving fine-grained periodic structure, which is desirable for quasi-periodic electrical signals because normality is expressed as stable cycle shape and cross-axis relationships rather than as a smooth low-frequency trend [37]. Temporal CNNs are also strong supervised baselines in time-series classification, suggesting the inductive bias is well matched to oscillatory structure [25].

Attention-based denoisers impose a different bias: they can explicitly relate distant time points, which can help when fault signatures are non-local (e.g., repeated intermittent bursts whose spacing carries information) [47, 49]. However, attention cost grows with window length, and in anomaly detection the inference budget and score stability often matter more than maximal expressivity. Therefore long-context attention is attractive only when it improves score calibration under drift rather than merely improving sample realism.

Conditioning is central for scoring stability. The denoiser must be informed of noise level, and in deployment-relevant settings it should also be able to separate "this is a different sensor/regime" from "this is a fault". Conditioning on sensor identity, mounting configuration, or coarse operating regime (when available) reduces spurious variance and makes normal-only calibration more defensible.

In diffusion models more broadly, conditioning can also be implemented via guidance mechanisms (classifier-guided or classifier-free), which can materially change the semantics and calibration of any downstream score [12, 23].

For this problem, preserving multi-axis structure is essential: HIF signatures are plausibly expressed as changes in axis asymmetry and cross-axis consistency. Architectures that process axes jointly can learn these relationships, but they must not erase asymmetry by construction. The practical objective is a model that preserves normal cross-axis structure while being sensitive to deviations in that structure.

Continuous (Gaussian) diffusion is the dominant formulation in the broader literature, and it motivates much of the terminology used throughout this report. For context, DDPM-style models corrupt x_0 with Gaussian noise over T steps and learn to denoise from x_t [22, 35]. In this project, the primary modelling and deployment analysis focuses on discrete diffusion on the lab dataset, while a Gaussian implementation is used mainly for benchmark-protocol comparison in Chapter 8.

2.5 Discrete diffusion for anomaly detection

2.5.1 Motivation: changing the geometry of mismatch

Discrete diffusion models offer an alternative formulation that is well suited to anomaly detection in structured, non-stationary magnetic-field time-series. The motivation here is calibration and interpretability under drift: discrete diffusion makes the representational and geometric assumptions of scoring explicit.

What goes wrong with Gaussian amplitude modelling. With Gaussian corruption, both reconstruction and likelihood-style scores inherit quadratic dependence on amplitude error. A reconstruction score has the form

$$s_{\mathrm{rec}}(x) = \frac{1}{LC} \sum_{i=1}^{LC} \left(x^{(i)} - \hat{x}_0^{(i)} \right)^2, \tag{2.1}$$

and a Gaussian negative log-likelihood term is

$$-\log p_\theta(x \mid z) = \frac{1}{2} \sum_{i=1}^{LC} \left(\frac{\left(x^{(i)} - \mu_\theta^{(i)} \right)^2}{\sigma_\theta^{2(i)}} + \log \sigma_\theta^{2(i)} \right) + c. \tag{2.2}$$

For magnetic-field sensing this can be a poor match: normal operation can be heavy-tailed and heteroskedastic (variance changes across cycle phase, axes, and operating regimes), so squared-error geometry can over-weight high-amplitude or high-variance regions relative to slower structural changes such as axis-asymmetry drift [42, 34, 17].

Why quantisation changes the geometry of "normal vs anomalous". Let $q(\cdot)$ map each amplitude to one of K bins. A token-level score at diffusion step t can be written as

$$s_{\mathrm{ce},t}(x) = -\frac{1}{LC} \sum_{i=1}^{LC} \log \pi_{\theta,t}^{(i)} \left(q\left(x^{(i)} \right) \right), \tag{2.3}$$

where $\pi_{\theta,t}^{(i)}$ is the model probability for token i at step t. Because $q(\cdot)$ collapses continuous amplitudes into finite bins, the score depends on bin disagreement rather than squared raw-amplitude distance. In practice this reduces direct leverage of isolated extreme amplitudes and emphasises persistent inconsistencies across positions and channels [18, 51, 3].

One key motivation for discrete diffusion in anomaly detection is the ability to compute an explicitly decomposed training objective and associated scoring terms. As in continuous diffusion, training is framed via a variational lower bound

$$\mathcal{L}_{\mathrm{VLB}}(x_0) = \mathbb{E}_{q(x_{1:T}|x_0)}\left[L_0 + \sum_{t=2}^{T} L_{t-1} + L_T\right], \qquad L_{t-1} = \mathrm{KL}(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)). \tag{2.4}$$

In the discrete setting, these KL terms have closed-form expressions because the forward posterior is analytic:

$$q(x_{t-1} = i \mid x_t = j, x_0 = k) = \frac{Q_t(i,j)\,\bar{Q}_{t-1}(k,i)}{\sum_{i'=0}^{K-1} Q_t(i',j)\,\bar{Q}_{t-1}(k,i')}, \qquad \bar{Q}_{t-1} = Q_1 Q_2 \cdots Q_{t-1}. \tag{2.5}$$

This gives transparent per-step contributions to an anomaly score and makes it straightforward to inspect whether mismatch is concentrated in early (coarse) or late (fine) denoising stages [3].

This kind of decomposable ELBO formulation is particularly valuable in safety-critical domains such as electrical infrastructure monitoring, where anomaly scores may inform operational decisions. In this dissertation's experiments, the reported "likelihood-style" metrics are Monte Carlo denoising-loss proxies (per-token cross-entropy over sampled timesteps) rather than full ELBO evaluations or true likelihood estimates; the ELBO decomposition remains the theoretical reference. Likelihood-style scores derived from variational objectives can be sensitive to modelling assumptions, noise schedules, or architectural choices, making it difficult to interpret absolute values or to compare scores across different conditions. Discrete diffusion mitigates this issue by grounding anomaly scores in well-defined per-step divergence terms under a fixed corruption process.

Discrete formulations also align naturally with practical constraints on signal representation. Real-world sensor data are already quantised during acquisition (finite ADC resolution) and effectively quantised by noise floors. Treating the signal as a continuous Gaussian variable can therefore add modelling burden without adding useful semantic resolution. A discrete representation acknowledges that many "differences" below a certain scale are not operationally meaningful and should not dominate the score.

From a scoring perspective, discrete diffusion offers advantages in stability and interpretability. The Markov structure provides clear semantics for how information is destroyed and reconstructed, and intermediate divergences can be analysed to determine whether a window fails at coarse structure (cycle-level consistency) or fine structure (local irregularities). This is directly relevant for distinguishing gradual HIF onset from benign high-frequency interference.

These properties make discrete diffusion an attractive foundation for anomaly detection in non-stationary, structured time-series measurements.

2.5.2 Discrete diffusion formulation

Discrete diffusion generalises the diffusion idea to a discrete state space. Let a quantised window be represented as $x_0 \in \{0, \ldots, K-1\}^{L \times C}$, where L is window length, C is the number of channels, and each element is a discrete bin index (matching the implementation's 0-indexed tokens; some literature uses $\{1, \ldots, K\}$ notation). Throughout this report, token-index notation follows the 0-indexed convention to match the implementation. The forward process is defined by a sequence of categorical transition kernels that progressively destroy information:

$$q(x_t \mid x_{t-1}) = \prod_{i=1}^{LC} \mathrm{Cat}\left(x_t^{(i)};\, Q_t\big(x_{t-1}^{(i)}, \cdot\big)\right), \tag{2.6}$$

where Q_t is a $K \times K$ transition matrix applied independently at each position. This factorised structure is not a limitation in the same way as assuming i.i.d. data; the dependence structure is captured by the learned reverse model, while the forward process is a controlled corruption used for training. Here $\mathrm{Cat}(\cdot;\, p)$ denotes a categorical distribution over K bins, $x_t^{(i)}$ is the i-th token in a flattened $(L \times C)$ index, and $Q_t(a, \cdot)$ denotes the row of Q_t corresponding to source bin a.

The choice of Q_t is a design decision that should be justified in terms of the signal. If bins have an ordinal amplitude meaning (e.g., uniform amplitude bins), then "local" transitions that preferentially move to nearby bins preserve coarse waveform shape under light corruption and only destroy fine detail at higher corruption. This matches the quasi-periodic structure of electrical signals: early denoising steps should recover cycle-level consistency, while later steps should refine small deviations. By contrast, uniform random replacement makes every mistake equally severe, which can make the task less aligned with physical amplitude continuity.

The reverse process is learned via a neural network parameterisation:

$$p_\theta(x_{t-1} \mid x_t) = \prod_{i=1}^{LC} \mathrm{Cat}\left(x_{t-1}^{(i)};\, \pi_\theta^{(i)}(x_t, t)\right), \tag{2.7}$$

where $\pi_\theta^{(i)}(\cdot)$ is the predicted categorical distribution for position i given the corrupted window x_t and timestep t. Unlike the forward process, which is fixed, the reverse model is where temporal and cross-axis structure is represented: π_θ can depend on the entire window through convolutional or attention-based architectures, allowing the model to infer which discrete values are consistent with the learned distribution of normal sequences.

Training optimises the decomposition above. For scoring, define per-step mismatch

$$\ell_t(x_0) = \mathbb{E}_{q(x_t \mid x_0)}\left[\mathrm{KL}(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t))\right], \quad t = 2, \ldots, T. \tag{2.8}$$

Then a likelihood-style anomaly score is $S_{\mathrm{KL}}(x_0) = \sum_{t=2}^{T} \ell_t(x_0)$ (or its Monte Carlo approximation using sampled timesteps). Because $q(\cdot)$ and Q_t are known, computing ℓ_t requires only the model's categorical predictions at sampled x_t, which is why stepwise diagnostics are tractable in D3PM-style models [3]. D3PM additionally uses an auxiliary cross-entropy term to improve optimisation and sample quality.

As with continuous diffusion, discrete diffusion models can also be conditioned or guided. While this dissertation focuses on robust scoring under normal-only calibration, it is useful to note that simple guidance mechanisms have been proposed specifically for discrete diffusion settings [40].

For anomaly detection, this stepwise decomposition is often more useful than the philosophical question of whether the resulting quantity is an "exact likelihood". What matters operationally is that the score is stable, comparable across windows, and interpretable as accumulated mismatch under a well-defined corruption-and-recovery process. In particular, stepwise KL terms provide a natural family of anomaly scores:

1. a total score (sum across timesteps) that measures overall mismatch;

2. a coarse-to-fine profile (where the score concentrates) that can help distinguish genuine structural anomalies from harmless noise;

3. axis-conditioned variants, where the score is computed per channel or emphasises cross-axis inconsistency.

The connection to quantisation now becomes explicit. Quantisation defines K and the geometry of the discrete space. In a heavy-tailed, heteroskedastic amplitude regime, quantisation also acts as a robustness prior: it compresses nuisance variability while preserving structured departures that persist across time. Coarse quantisation can erase subtle early-stage fault signatures; overly fine quantisation can make the score hypersensitive to benign calibration drift. Therefore K is not merely a representational choice but a modelling commitment about what deviations count as operationally meaningful.

Finally, discrete diffusion provides a clean route to "masking-style" corruption via absorbing states. In such designs, the forward process increasingly replaces values with a special mask token, and the reverse model learns to infill plausible values. For anomaly detection, this reframes reconstruction as projection onto the learned normal manifold: anomalous structure is replaced (or fails to be recovered) because it is not consistent with learned normality. For quasi-periodic magnetic-field sensing, this provides an operationally intelligible interpretation: if the model must make many bin-level changes across the window (especially in axis-consistency patterns), the window is more plausibly a fault-related deviation than a harmless transient.

2.6 Anomaly detection using diffusion-based models

The following subsections give a high-level view of scoring concepts and related literature. Concrete scoring methods and evaluation choices for this project are defined in Chapters 6 and 7. The focus here is diffusion as an *operational scoring tool*: turn a trained window-level model into a scalar score that can be kept calibrated under drift using normal-only data.

2.6.1 Scoring anomalies in generative models

A central challenge in applying generative models to magnetic-field HIF detection is defining a scalar score that tracks *operational deviation* rather than nuisance variability. In diffusion-style detectors, three broad classes of scoring approaches are common:

- **Likelihood-style surrogates:** scores derived from the training objective (e.g., ELBO terms or denoising-loss proxies), treating anomalies as low-probability samples.
- **Reconstruction-based scores:** residuals after partially corrupting an observation and projecting it back toward the data manifold.
- **Stepwise / hybrid scores:** aggregations of denoising errors or divergence terms across timesteps (and potentially across channels), providing a coarse-to-fine mismatch profile.

For a window x, common instantiated scores are:

$$s_{\text{rec}}(x) = \frac{1}{LC} \left\| x - \hat{x}_0(x_{t_{\text{rec}}}) \right\|_2^2, \tag{2.9}$$

$$s_{\text{like}}(x) = \frac{1}{M} \sum_{m=1}^{M} \ell_{t_m}(x), \qquad t_m \sim \text{Unif}\{2, \ldots, T\}, \tag{2.10}$$

$$s_{\text{step}}(x) = \sum_{t=2}^{T} w_t \, \ell_t(x), \tag{2.11}$$

where $\ell_t(x)$ is a per-step mismatch term (KL or a per-step token cross-entropy proxy) and w_t controls coarse-vs-fine emphasis [22, 3].

Likelihood-style scores are principled but can be sensitive to modelling choices (noise schedules, parameterisation, capacity) and may emphasise low-level statistics. Reconstruction residuals are intuitive but can be metric-sensitive for oscillatory signals (e.g., phase shifts) and can fail if the model generalises too broadly. Stepwise decompositions improve diagnostics by revealing where along the denoising trajectory mismatch concentrates, and in discrete diffusion they naturally produce bin-level error maps that align better with heavy-tailed, heteroskedastic amplitude regimes than squared-error residuals.

2.6.2 Sliding windows for sequential anomaly detection

Most diffusion models operate on fixed-length sequences, so streaming magnetic-field data are segmented into fixed-length windows (with optional overlap). For high-impedance faults, window length sets the evidence horizon: short windows mostly capture cycle-level shape, while longer windows can capture slower axis-asymmetry changes and staged fault evolution [1, 9, 42]. Stride and overlap then control the update rate of the score and the correlation of successive values, which in turn determines how a fixed threshold maps to an alert rate.

In deployment-relevant evaluation, window scores should therefore be interpreted through a causal alerting policy (smoothing, persistence, cooldown). Without that policy, a detector can appear to perform well under AUROC/PR-AUC while still producing an unacceptable number of false alerts per day.

2.6.3 Evaluation criteria under imbalance and drift

For this problem class, evaluation is not only about reconstruction quality. A deployable detector needs useful ranking behaviour under heavy class imbalance and stable operating points under drift. This motivates a split between *fidelity diagnostics* (does the model represent normal structure?) and *detection diagnostics* (does score/threshold behaviour stay operationally useful?).

Pointwise versus event-based evaluation

Faults are temporally extended events, not i.i.d. samples. Pointwise metrics are useful for diagnostics, but event-level outcomes are often closer to operational requirements. A practical event rule is: an event is detected if the score crosses threshold at least once (optionally with persistence) within a tolerance window after onset.

This distinction makes latency visible. A detector can look strong pointwise yet still be operationally weak if alerts arrive too late in the event, or if alerting logic suppresses true detections.

Ranking metrics under class imbalance

Because positive windows are sparse, PR-based metrics are usually more informative than ROC alone [11, 39]. ROC can remain optimistic at low prevalence even when precision is poor. In this regime, AUROC and PR-AUC should be reported together, and PR-AUC should be interpreted relative to prevalence (random-ranking baseline).

Where ROC-style summaries are reported, operating-point context is still needed [13]: in deployment, acceptable false-positive behaviour is usually specified in time units (for example, false alerts per day), not as a raw false-positive fraction.

Alert-rate metrics and persistence

A detector is only usable if its alert rate is tolerable. For sliding-window pipelines, alert rate depends jointly on threshold and post-processing logic (smoothing, persistence, cooldown). Reporting both threshold-free ranking metrics and thresholded alert-rate metrics clarifies whether gains come from score separability or from downstream heuristics.

This is particularly important for HIF detection, where weak early signatures can force a trade-off between sensitivity and alert burden. Persistence can suppress nuisance spikes, but it also increases delay and can reduce detected-event fraction.

Threshold calibration and drift

Oracle thresholds (chosen on evaluation labels) are useful diagnostic upper bounds but are not deployment-operating points. Deployment calibration should use presumed-normal data and an explicit false-alarm budget, then evaluate how stable that operating point remains under distribution shift [14, 50].

When the threshold required for a fixed alert rate drifts substantially across sensors, days, or regimes, this indicates score-calibration fragility rather than a simple tuning problem.

Protocol validity

Evaluation is sensitive to leakage and contamination. If unlabelled faults enter training data, anomaly scores can be diluted. If overlapping windows from the same long recording are split across train and test, apparent performance can be inflated. Session/file-level split discipline and explicit reporting of split strategy are therefore core parts of anomaly-detection validity.

2.6.4 Related work: diffusion models for time-series modelling and anomaly detection

Diffusion models were initially popularised in vision, but a growing body of work explores diffusion for time-series. Much of that literature is oriented toward forecasting and sample realism. For example, TimeGrad [36] uses diffusion to model predictive distributions in an autoregressive forecasting pipeline. More recent work explores hybrid strategies such as retrieval-augmented diffusion to improve robustness and long-range behaviour in forecasting settings [30]. This is a strong fit when the objective is calibrated prediction under a stable regime.

For high-impedance fault detection under drift, the objective is different: the deliverable is a stable anomaly score under a false-alarm budget, not the best forecast. Forecasting-optimised pipelines can fail to transfer because they reward average predictive accuracy and may treat rare regime changes as noise to average out. The key question for this report is therefore not whether diffusion generates plausible sequences, but whether its scores stay stable and useful for quasi-periodic magnetic-field sensing.

Diffusion-based anomaly detection has developed rapidly, largely within vision, where reconstruction-based paradigms are common: a model is trained on normal data and used to "repair" anomalous inputs, with the reconstruction residual used to localise or score anomalies. AnoDDPM [54], DiffusionAD [58], GLAD [57], and subsequent variants such as diffusion-ODE scoring or masked-posterior sampling [24, 53] illustrate the breadth of this approach. Recent work extends this thread with binary diffusion for anomaly detection, industrial robustness, and broader diffusion-based anomaly detection overviews [52, 8, 7, 10].

This is directionally relevant, but magnetic-field time-series introduce additional pitfalls: naive residual metrics are easily dominated by harmless phase and amplitude variability, and score calibration under drift is often the main practical failure mode.

OOD detection context (high level). OOD detection has an extensive literature, spanning confidence- and post-hoc methods (e.g., maximum softmax probability and ODIN) [20, 29], representation-space scoring (e.g., Mahalanobis distance) [27], and modern energy-based scores [31]. These methods are attractive for different reasons: temperature scaling and input perturbation (ODIN) can sharpen separability when models are well calibrated; Mahalanobis distance leverages class-conditional feature distributions and can be sample-efficient; and energy-based scores often provide more stable thresholds than raw softmax confidence under shift. Recent surveys highlight that real deployments often care less about a single benchmark AUC and more about robustness under shift and clear problem definitions [33].

Diffusion-specific OOD detection methods exploit diffusion-native signals beyond raw likelihood, such as denoising error across noise levels [17] or geometry along the denoising trajectory (diffusion-path structure) [21]. This is directly aligned with the stance of this report: treat the corruption-and-repair trajectory as a source of diagnostic, potentially more stable scores.

For time-series specifically, benchmark-driven work often evaluates diffusion detectors on anomaly types that are visually obvious, i.i.d.-like, or labelled densely. That setting is informative but incomplete for distribution monitoring: HIF signatures can be subtle and temporally extended, labels are scarce, and calibration under drift is frequently the dominant deployment constraint. Therefore, this report treats the benchmark literature as motivation for feasibility, but not as a sufficient argument for deployability.

This work differs from much of the above in two ways. First, it focuses on non-invasive magnetic-field monitoring in distribution systems, where dominant structure is quasi-periodic and axis-dependent and where fault signatures are often subtle and temporally extended. Second, it prioritises score stability and interpretability under non-stationarity, which motivates discrete diffusion and stepwise scoring diagnostics over pure reconstruction quality. The goal is selective reconstruction: preserve normal structure while failing in a diagnostically useful way on anomalous structure.

2.7 Operational constraints for anomaly detection systems

Deployment constraints shape both model and score design. A diffusion-based detector has to do more than look good offline: it must run within latency and compute limits and keep operator trust by controlling false alerts. This is especially important for HIFs, which evolve over longer timescales but still require timely, credible alerts.

2.7.1 Latency and temporal context

Detection latency is determined jointly by window length, window stride, and any persistence criteria used to suppress spurious triggers. Short windows and small strides enable faster response but reduce context and increase score variance; long windows stabilise scores and capture gradual evolution but inherently delay detection. Persistence rules (e.g., requiring the score to exceed a threshold for multiple consecutive windows) improve robustness at the cost of further delay [1, 19, 48]. The system design must therefore select a latency budget and tune windowing and alert logic to meet it.

For high-impedance faults, early-stage deviations may be subtle and distributed. In this regime, the system benefits from enough context to detect changes in symmetry, harmonic structure, or gradual drift rather than relying on instantaneous outliers. This motivates windowing choices that capture multiple AC cycles and enough temporal span to observe evolving structure, paired with alert logic that balances sensitivity and stability.

2.7.2 Compute and throughput constraints

Diffusion inference typically involves multiple denoising steps per window, making compute cost a primary deployment consideration. Even when training is performed centrally, inference may need to run near the sensor (edge) or in a constrained central pipeline processing many sensors. Compute cost scales with window length, model size, and the number of denoising steps used at inference time.

Several practical strategies reduce inference cost without abandoning diffusion modelling. The number of denoising steps can be reduced through step subsampling schedules (e.g., DDIM-style updates) [45] or through fast ODE-based samplers [32], and inference can be restricted to a limited number of candidate windows identified by lightweight pre-filters. Model architectures can be chosen to emphasise efficient temporal convolutions over expensive attention operations at long window lengths. In addition, scoring can be designed to use quantities already computed during denoising, avoiding additional passes through the model.

These considerations favour designs where the diffusion model produces a small set of interpretable scores per window that can be thresholded and aggregated efficiently.

2.7.3 Robustness to drift and update cadence

Distribution monitoring occurs in non-stationary environments. Sensor drift, mounting variability, changing load patterns, and network reconfiguration shift the normal operating distribution over time. A static model and static threshold can therefore degrade even if the underlying fault mechanisms remain unchanged.

Operationally, this suggests the need for drift-aware processes: monitoring the score distribution under presumed-normal periods, flagging sustained score inflation as potential drift rather than fault, and defining a safe update cadence for recalibration or retraining. Because labelled faults are scarce, updates must be designed to avoid inadvertently "learning away" anomalies; retraining on recent data may require careful filtering of candidate normal periods or human-in-the-loop validation.

2.7.4 Interpretability, diagnostics, and trust

In safety-relevant infrastructure, operators need to understand why an alert was raised. Diffusion models offer a useful advantage here: reconstruction outputs and stepwise score decompositions can provide diagnostic artefacts. Visualising the original versus reconstructed waveform, comparing axis-wise behaviour, and inspecting how errors concentrate in particular frequency bands can help differentiate genuine fault-like behaviour from benign interference.

To be operationally useful, scores need to map to inspectable quantities (for example, per-channel residuals or per-step KL terms) so alerts can be reviewed and false positives can be debugged.

2.7.5 False positives and operational cost

In monitoring systems, false positives are a primary operational failure mode. Excessive false alarms lead to alert fatigue, degrade trust in the system, and can cause operators to ignore real events [19, 48]. As a result, models are often tuned to strict false alarm budgets, and improvements in sensitivity are only valuable if they do not substantially increase nuisance alerts.

This prioritisation affects modelling choices. A detector that is marginally more sensitive but poorly calibrated under drift may be worse in practice than a slightly less sensitive detector with stable, controllable alert rates. Evaluation therefore must report false alarms in operational units (e.g., per day) and test across plausible distribution shifts rather than only on a stationary test set.

2.7.6 Integration with downstream decision logic

Finally, diffusion-based scoring is typically one component of a broader decision pipeline. Practical systems may combine model scores with rule-based checks (for example, requiring axis asymmetry to accompany a high score) or with escalation logic that distinguishes "warning" from "critical" alerts based on persistence and magnitude. This layered design reflects a pragmatic separation of responsibilities: the diffusion model learns the distribution of normal behaviour, while downstream logic implements conservative decision rules aligned with operational priorities.

Part II

System, experiments, and deployment

Chapter 3

Problem formulation and system overview

This report considers one-class anomaly detection for low-voltage distribution monitoring using high-sample-rate magnetic-field time-series. The sensing modality is non-invasive: magnetic sensors observe field variations induced by line currents without requiring direct electrical contact [41, 42]. The electrical context and sensing assumptions are introduced in Chapter 2, and the laboratory dataset and channel definitions used in experiments are described in Chapter 4.

3.1 Problem formulation: one-class sequential anomaly detection

The operational framing and assumptions are established in Section 2.1.1. This chapter records the concrete implementation used in the project.

Let $x(t)$ denote the measured magnetic-field stream, with $x(t) \in \mathbb{R}^C$. The online interface to the model is a sequence of fixed-length windows

$$x^{(n)} = x_{:,\,nS:nS+L-1} \in \mathbb{R}^{C\times L},$$

where L is the window length and S is the stride in samples. Training uses windows from presumed-normal periods, and inference produces a score $s(x^{(n)})$ for each window.

In this report, threshold calibration uses normal-only data and is not tuned on test labels. Thresholded decisions are then formed by combining a fixed threshold with causal persistence/cooldown policy, matching the deployment objective of low false-alarm burden under drift.

3.2 System overview

Figure 3.1 conceptually summarises the end-to-end pipeline used throughout this report. First, raw experiment recordings are loaded from MATLAB files and preprocessed into normalised multi-channel windows (Chapter 4). Second, a diffusion model is trained on windows from presumed-normal periods (Chapters 2 and 5). Third, at inference time the trained model assigns anomaly scores using reconstruction-based criteria and/or denoising-loss proxies (Chapter 6). Finally, scores are aggregated across time into event-level decisions and evaluated using both reconstruction fidelity metrics and detection performance measures (Chapters 7 and 8).

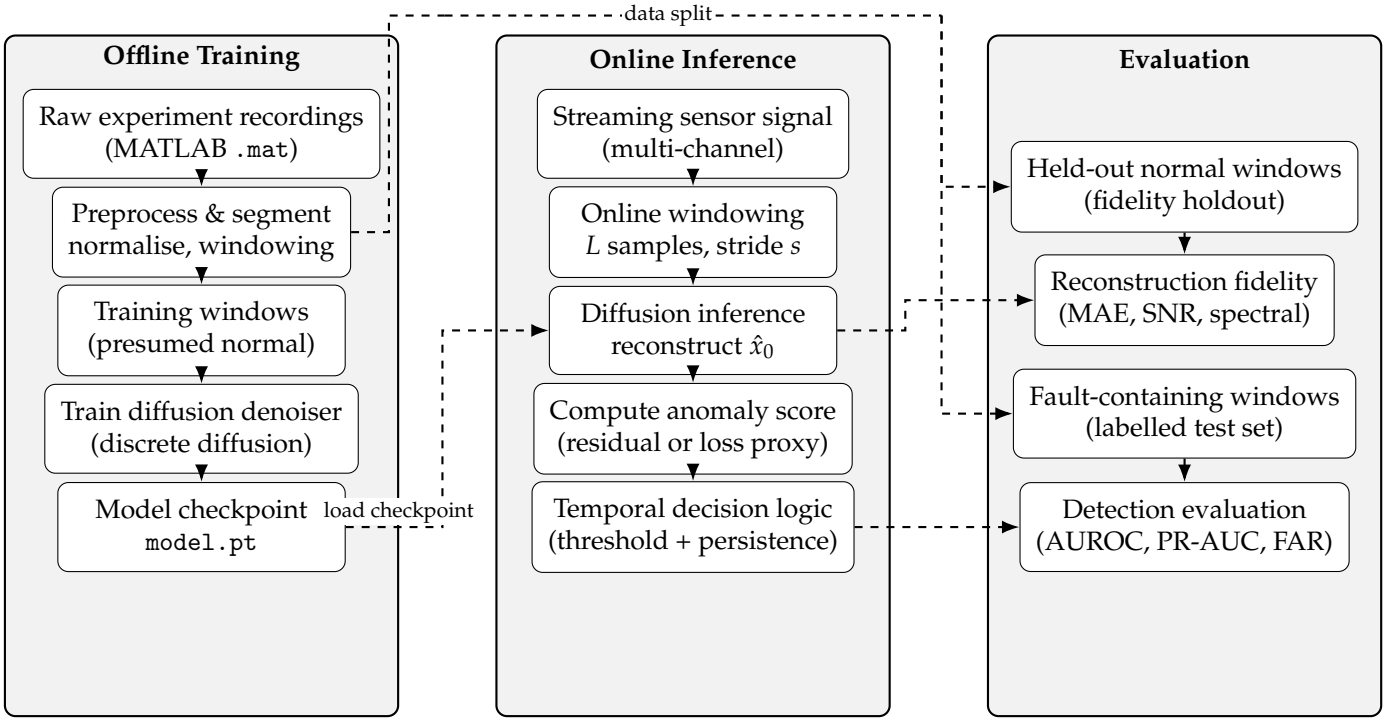


Figure 3.1: End-to-end pipeline: offline preprocessing and diffusion training on presumed-normal windows, online window-level scoring with temporal decision logic, and evaluation using both reconstruction fidelity and detection metrics.

Chapter 4

Data and preprocessing

The laboratory magnetic-field dataset and preprocessing pipeline are documented here, including the transformation from raw recordings to fixed-length windows for diffusion modelling and anomaly detection. The implementation is in `scripts/train_eval.py` and `src/eds_diffusion/lab_data.py`.

4.1 Laboratory dataset format

Each experiment is stored as a MATLAB `.mat` file containing a time column and multiple sensor channels. During loading, the time column is not used as a model input: it is dropped and only the sensor channels are retained.

4.1.1 Channel definitions ($C = 16$)

In the provided laboratory files, the primary array `BHTWTdataS100` has 17 columns: a timestamp column followed by 16 feature channels. After dropping the timestamp, the model input channels are taken in the order they appear in `BHTWTdataS100`: Here $\mathcal{H}(\cdot)$ denotes the Hilbert transform. The accompanying dataset notes describe the 16 non-time channels only at a high level as "2x sensor data and HT and DWT outputs" and do not provide per-column headers for the wavelet outputs; accordingly, this report refers to the wavelet-derived columns generically as W_1 through W_{10} while preserving their fixed ordering throughout training and evaluation.

For the laboratory data used here, the sampling rate is treated as known and fixed, which allows reporting latency and false-alarm rates in time units.

The recordings originate from a purpose-built low-voltage (400 V) physical simulation facility for high-impedance fault experiments, with current and magnetic-field measurements captured for subsequent analysis [41, 42].

The resulting sample presented to the model is a windowed tensor with shape (C, L), where C is the number of channels and L is the window length in samples.

4.2 Train/test separation

The dataset loader supports two common directory layouts:

- Training data in `Training Datasets/` (or `Training/`).
- Testing data in `Testing Datasets/`, `Testing - Linear/`, or `Testing - Nonlinear/`.

Col.	Symbol	Description
1	$B^{(1)}(t)$	Raw magnetic-field sensor channel 1.
2	$\left\|\mathcal{H}(B^{(1)})(t)\right\|$	Hilbert-transform envelope magnitude of channel 1.
3	$\angle\mathcal{H}(B^{(1)})(t)$	Hilbert-transform instantaneous phase of channel 1 (stored approximately in degrees in the source files, with values in a ± 180 range).
4	$B^{(2)}(t)$	Raw magnetic-field sensor channel 2.
5	$\left\|\mathcal{H}(B^{(2)})(t)\right\|$	Hilbert-transform envelope magnitude of channel 2.
6	$\angle\mathcal{H}(B^{(2)})(t)$	Hilbert-transform instantaneous phase of channel 2 (stored approximately in degrees).
7	$W_1(t)$	Wavelet-derived output channel 1.
8	$W_2(t)$	Wavelet-derived output channel 2.
9	$W_3(t)$	Wavelet-derived output channel 3.
10	$W_4(t)$	Wavelet-derived output channel 4.
11	$W_5(t)$	Wavelet-derived output channel 5.
12	$W_6(t)$	Wavelet-derived output channel 6.
13	$W_7(t)$	Wavelet-derived output channel 7.
14	$W_8(t)$	Wavelet-derived output channel 8.
15	$W_9(t)$	Wavelet-derived output channel 9.
16	$W_{10}(t)$	Wavelet-derived output channel 10.

Table 4.1: Channel definitions for the 16 non-time columns of `BHTWTdataS100` (in order, after dropping the timestamp).

The `mode=train` split draws windows from the training directories, and `mode=test` draws windows from the testing directories. Normalisation statistics are computed from the training files only to avoid leakage.

4.3 Windowing

Let $x \in \mathbb{R}^{T\times(C+1)}$ denote a raw recording with time in the first column. For each file, contiguous windows are extracted:

- Window length L is set by `data.seq_len`.
- Window stride defaults to L (non-overlapping windows), but can be configured.

For each window, the model input is $x_{\text{win}} \in \mathbb{R}^{C\times L}$ (channels-by-time).

4.4 Fault labels and masks

When label annotations are present in a `.mat` file, the loader constructs a binary mask $m \in \{0,1\}^L$ by reading the provided label information (e.g., start/end time intervals in a `LabelsS100/Labels100` structure, or a dense 0/1 label vector) and slicing it to the extracted window. When no labels are present, the mask is all zeros. Filenames and directory layout are used for dataset organisation (train/test splits and reporting), but they do not gate whether labels are parsed.

The dataset returns (x, x_0, m) where x and x_0 are identical in the current implementation and have shape (C, L), and m has shape $(L,)$. This supports a one-class training setup (train on normal windows) with supervised evaluation on fault windows.

4.4.1 Label integrity / labelling validation

To validate the annotation quality, the dataset was scanned automatically with the inspection script (`scripts/inspect_lab_data.py`) to detect short or otherwise suspicious fault annotations and to cross-check the fault-current channel for the expected 50 Hz onset signature. No flagged events were found in the testing datasets after these checks.

Figure 4.1 summarises the distribution of fault durations computed from contiguous label masks (start/end intervals or dense 0/1 vectors) and using the per-file timestamps from the first column of the `BHTWTdataS100` dataset. When timestamps were not available, durations were inferred from a default sample rate of 1000 Hz. A CSV summary of the aggregated durations is available at `runs/label_validation/combined_fault_duration_by_type.csv`.

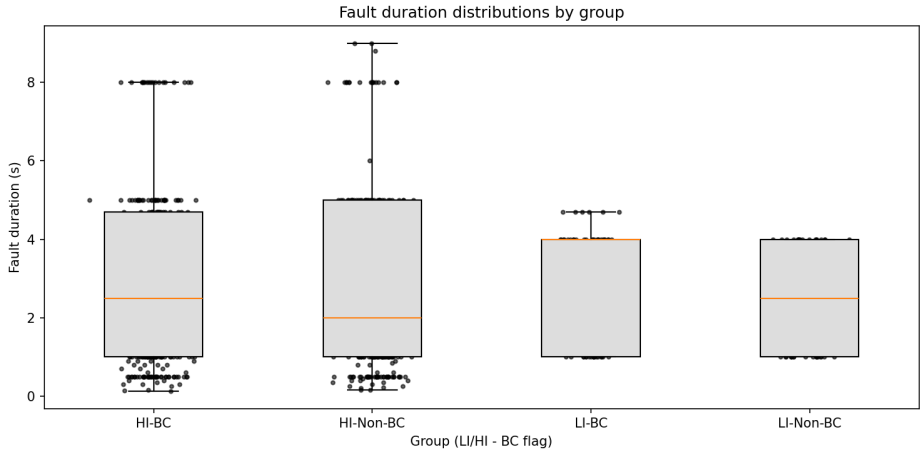


Figure 4.1: Distribution of fault durations (seconds) by group: LI vs HI and BC vs Non-BC. Durations are computed from contiguous label masks and file timestamps; sample sizes and medians are reported in the figure caption in the main text.

4.5 Normalisation

All normalisation is applied after dropping the time column and before transposing to (C, L). Two strategies are supported:

- **Min–max:** per-channel global min/max computed from training files and mapped affinely to $[-1, 1]$.
- **Standard deviation scaling (default):** per-channel global mean μ and standard deviation σ computed from training files, followed by

$$\tilde{x} = \text{clip}\left(\frac{x - \mu}{k\,\sigma},\, -1,\, 1\right),$$

  where k corresponds to `data.normalization_std_clip`.

The standard-deviation strategy keeps values within a fixed bounded range, which matches the assumed input range for quantisation in the discrete diffusion setting.

4.6 Quantisation for discrete diffusion

Discrete diffusion models require mapping a normalised waveform $\tilde{x} \in [-1, 1]$ to integer tokens. In the implementation, quantisation is handled by `AmplitudeQuantizer` and is configured via:

- number of bins K (`diffusion.num_bins`);
- clipping range $(v_{\min}, v_{\max})$ (`diffusion.value_range`, typically $[-1, 1]$);
- quantiser strategy (`diffusion.quantizer_strategy` in {`uniform`, `cdf`}).

All quantisation paths clamp inputs to $(v_{\min}, v_{\max})$ to handle outliers consistently. After normalisation (Section 4), this corresponds to clipping extreme values into the supported representational range.

4.6.1 Uniform binning

For `quantizer_strategy=uniform`, the range $(v_{\min}, v_{\max})$ is partitioned into K equal-width bins with edges

$$e_i = v_{\min} + i \, \frac{v_{\max} - v_{\min}}{K}, \quad i = 0, \ldots, K,$$

and bin centres $c_i = \frac{1}{2}(e_i + e_{i+1})$. Quantisation maps a sample to the corresponding bin index in $\{0, \ldots, K-1\}$, and dequantisation maps a token back to its bin centre.

4.6.2 CDF/quantile binning (training-fit)

For `quantizer_strategy=cdf`, bin edges are fit on the *training split only* before training begins. The code collects up to `diffusion.cdf_max_samples_per_channel` scalar samples per channel from training windows and computes quantile edges so that each bin contains approximately equal probability mass under the training distribution. Concretely, for each channel $c \in \{1, \ldots, C\}$, let F_c denote the empirical CDF of the collected training samples (after clamping to $(v_{\min}, v_{\max})$). Define quantile levels

$$\tau_i = \frac{i}{K}, \quad i = 0, \ldots, K,$$

and corresponding (per-channel) bin edges

$$e_i^{(c)} = F_c^{-1}(\tau_i), \quad i = 0, \ldots, K.$$

Given edges $\{e_i^{(c)}\}_{i=0}^{K}$, quantisation for a scalar sample x in channel c is then the bin index

$$q^{(c)}(x) = \max\{i \in \{0, \ldots, K-1\} : e_i^{(c)} \le x < e_{i+1}^{(c)}\},$$

with endpoint conventions applied so that $x = v_{\max}$ maps to $K - 1$. Dequantisation maps a token back to a representative value, e.g.

$$\hat{x} = \tfrac{1}{2}\left(e^{(c)}_{q^{(c)}(x)} + e^{(c)}_{q^{(c)}(x)+1}\right).$$

In the current implementation, these edges are fit *per channel* (separate edges for each of the C channels) and then reused for all windows in training and evaluation. Edges are clamped to $(v_{\min}, v_{\max})$ and adjusted to be strictly increasing; if the quantile edges become degenerate, the implementation falls back to uniform edges.

Practical implication. When the amplitude distribution is highly non-uniform (e.g., strongly peaked or with long tails), uniform bins can waste resolution in rarely visited tail regions while under-resolving dense regions near the mode. This can increase quantisation error for typical normal behaviour and can make discrete reconstructions "jittery" when small amplitude changes flip between adjacent bins. CDF/quantile bins allocate more bins where the signal spends most of its time, which can improve reconstruction fidelity and stabilise token-based scores.

The implications of K and binning strategy for reconstruction fidelity and anomaly score stability are explored in Chapter 8.

Chapter 5

Discrete diffusion model for time-series

The discrete diffusion model used in this report is implemented in `src/eds_diffusion/diffusion.py` and `src/eds_diffusion/models.py`. It represents fixed-length multi-channel magnetic-field windows by quantising amplitudes into a finite token vocabulary and learning a reverse denoising process over the resulting categorical sequences.

5.1 Tokenisation of time-series windows

Given a normalised window $x_0 \in [-1, 1]^{C \times L}$, each sample is mapped to an integer token via uniform amplitude quantisation (Chapter 4). This produces a discrete representation

$$z_0 \in \{0, \dots, K-1\}^{C \times L}, \tag{5.1}$$

where K is the number of bins.

In the implementation used in this report, tokens retain their natural (unflattened) (C, L) layout. In code, this corresponds to treating tokens as a `(B, C, L)` tensor (channels preserved).

5.2 Forward (corruption) process

The forward process is a simple categorical corruption process that replaces tokens with random values. Let $t \in \{0, \dots, T-1\}$ be the diffusion timestep. Define $\bar{\alpha}_t$ as a cumulative "keep" probability derived from a standard beta schedule (linear or cosine) as in the diffusion literature [22].

For each token position, the forward process samples

$$z_t = \begin{cases} z_0, & \text{with probability } \bar{\alpha}_t, \\ U, & \text{with probability } 1 - \bar{\alpha}_t, \end{cases} \tag{5.2}$$

where $U \sim \text{Uniform}(\{0, \dots, K-1\})$.

This "random replacement" corruption is closely related to the uniform categorical diffusion processes explored in discrete diffusion work (e.g., D3PM [3]), and resembles masking-style corruption in that information is progressively destroyed as t increases.

5.2.1 Biased timestep sampling

Standard diffusion training samples timesteps uniformly: $t \sim \text{Uniform}(\{0, \ldots, T-1\})$. For anomaly detection, low-noise timesteps can be more informative for subtle deviations because they preserve fine-grained structure. We therefore implement optional biased sampling strategies (e.g., *early bias* $p(t) \propto \text{linspace}(1.0, 0.1, T)$) as a pragmatic capacity-allocation heuristic. This can improve sensitivity at similar compute, but it is not guaranteed: emphasising some timesteps can trade off performance at others, and an importance-sampling interpretation requires appropriate reweighting.

5.3 Denoising model

The denoiser is implemented as `TokenEmbeddingDenoiser`. It consists of:

- An embedding layer mapping each token to a learned vector in $\mathbb{R}^D$.
- A 1D residual U-Net backbone (`Denoiser1D`) operating on the embedded sequence, conditioned on timestep t via a sinusoidal embedding.
- A 1×1 convolutional classifier projecting features back to categorical logits over K tokens per position.

Using the unflattened (C, L) token layout, the network treats channels as separate input "channels" to the 1D U-Net.

5.4 Training objective and optimisation

During training, a timestep t is sampled for each example in a mini-batch (uniform by default, or from a configured biased distribution), the forward process constructs z_t, and the model predicts logits $\ell_\theta(z_t, t)$ for the clean tokens. The training loss is the per-position cross-entropy:

$$\mathcal{L}(\theta) = \mathbb{E}_{z_0,t}\left[\text{CE}\big(\ell_\theta(z_t, t),\ z_0\big)\right]. \tag{5.3}$$

Optimisation follows the shared training loop:

- Adam optimiser with cosine annealing learning-rate schedule.
- Gradient clipping (global norm) to improve stability.
- Exponential moving average (EMA) of parameters.

5.5 Extensions (optional)

The following training extensions are implemented in the codebase but were not evaluated in the main rerun unless explicitly stated.

5.5.1 Contrastive learning with pseudo-anomalies

To prevent the model from assigning high likelihood to out-of-distribution samples—a common failure mode for generative anomaly detectors [34]—we implement contrastive learning. The model is explicitly trained to distinguish normal samples from synthetically corrupted versions (pseudo-anomalies). We implement five domain-specific corruption strategies detailed in Chapter 7: frequency injection, amplitude scaling, channel decorrelation, temporal shuffling, and quantile outliers.

The training objective is augmented with a margin-based contrastive term:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{recon}}(\mathbf{x}) + \lambda \, \max\Big(0, \; m - \big(\mathcal{L}_{\text{recon}}(\tilde{\mathbf{x}}) - \mathcal{L}_{\text{recon}}(\mathbf{x})\big)\Big). \tag{5.4}$$

where $\tilde{\mathbf{x}}$ is a pseudo-anomaly, $\mathcal{L}_{\text{recon}}(\cdot)$ is the chosen reconstruction/denoising loss used as a base score, $\lambda > 0$ weights the contrastive term, and $m > 0$ is the margin that enforces separation between normal and pseudo-anomalous losses. This forces the model to learn a sharper decision boundary, effectively teaching it what to reject [46, 5, 16].

The schedule hyperparameters T and schedule type (linear/cosine) are set consistently across experiments for reproducibility.

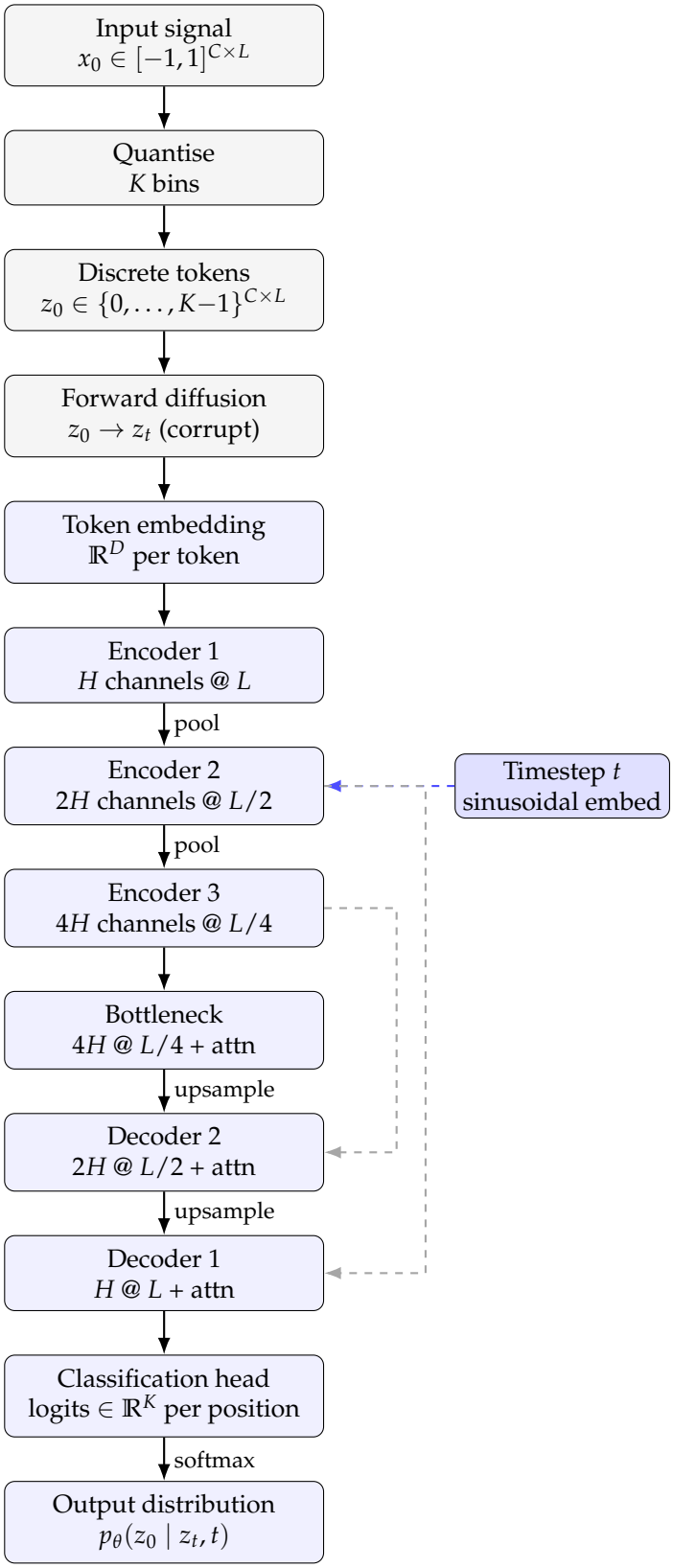


Figure 5.1: Discrete denoiser architecture: input signals are quantised to tokens, corrupted via forward diffusion, embedded, and processed by a 1D residual U-Net conditioned on timestep t. Skip connections preserve spatial information across encoder-decoder levels. Self-attention layers capture long-range dependencies in the bottleneck and decoder. The output is a categorical distribution over K bins for each position.

Chapter 6

Anomaly detection and scoring methods

A trained diffusion model is converted here into anomaly scores for fixed-length windows, with aggregation and thresholding definitions used for evaluation. The scoring procedures correspond to `src/eds_diffusion/metrics.py` and the evaluation logic in `scripts/train_eval.py`.

6.1 Problem setting

Following the one-class setup in Chapter 3, the diffusion model is trained on windows drawn from the normal operating regime (training files). At test time, the model is evaluated on a mixture of normal and fault windows, where fault intervals are provided as binary masks $m \in \{0,1\}^L$ (Chapter 4).

Let $x \in \mathbb{R}^{C\times L}$ denote a test window and $m \in \{0,1\}^L$ its per-timestep fault mask.

6.2 Reconstruction-based anomaly score

The reconstruction score measures how well the model can reconstruct an input after it has been partially corrupted by the forward process [54, 58]. For a fixed timestep t_{rec}, the pipeline is:

1. Corrupt x to obtain $x_{t_{\text{rec}}}$.
2. Apply the denoiser to estimate $\hat{x}_0$.
3. Score each timestep by squared error.

The model predicts clean-token logits and $\hat{x}_0$ is obtained by argmax over tokens followed by dequantisation.

The per-channel residual is

$$r_c(t) = \left(\hat{x}_{0,c}(t) - x_c(t)\right)^2,$$

and for multi-channel lab data, the implementation aggregates to a single per-timestep score by taking the maximum across channels:

$$s(t) = \max_{c\in\{1,\dots,C\}} r_c(t).$$

This design choice reflects the operational requirement that an anomaly in *any* channel should raise an alert, rather than requiring anomalies to be diffuse across all channels. Max aggregation preserves sensitivity to localised channel-specific faults that would otherwise be diluted by averaging over many normal channels.

6.3 Likelihood / ELBO-proxy anomaly score

The likelihood-style score reported in this project is an ELBO-motivated proxy computed by averaging per-timestep (or per-token) denoising losses over randomly sampled diffusion timesteps [34, 17].

Exact computation of the marginal likelihood $p(x_0)$ requires summing over all diffusion paths and is therefore intractable for realistic sequence lengths. In discrete diffusion, the ELBO decomposition is analytically defined and its per-step terms are tractable to estimate under the known forward process. In this work, the reported metrics use a Monte Carlo denoising-loss proxy (token cross-entropy at sampled timesteps), rather than a full ELBO evaluation or a true likelihood estimate.

6.3.1 Discrete diffusion

For discrete diffusion, the model outputs categorical logits over tokens. The score is the average per-token cross-entropy loss between predicted logits and the clean tokens, again averaged over random timesteps. This yields a per-position loss map that can be reshaped back to (C, L), which is then aggregated across channels using the same max-pooling strategy as reconstruction scores. Accordingly, when later chapters refer to "per-step decomposition" in the reported runs, this means per-timestep denoising-loss contributions from the proxy path.

6.4 Extensions (optional)

The following scoring extensions are available but were not evaluated in the main rerun.

6.4.1 Gradient-based sensitivity scoring

In addition to static errors, we introduce gradient-based scoring to measure the sensitivity of the reconstruction loss to input perturbations. The score is defined as:

$$s_{\text{grad}}(\mathbf{x}) = \|\nabla_{\mathbf{x}} \mathcal{L}_{\text{recon}}(\mathbf{x}, t; \theta)\| \tag{6.1}$$

where $\mathcal{L}_{\text{recon}}$ is the reconstruction loss and $\|\cdot\|$ denotes the ℓ_2 norm over the input tensor entries. This captures how hard the model works to denoise the input; anomalies require larger gradient corrections. For discrete diffusion, we approximate this via finite differences with a small scalar step $\epsilon > 0$ and $f(\mathbf{x})$ denoting the scalar loss being differentiated:

$$\nabla_{\mathbf{x}} f(\mathbf{x}) \approx \frac{f(\mathbf{x} + \epsilon) - f(\mathbf{x} - \epsilon)}{2\epsilon} \tag{6.2}$$

This approach is grounded in out-of-distribution detection methods that use score-norm signals [27, 38].

6.4.2 Inpainting-based anomaly amplification

We also implement scoring via *inpainting*. A sliding window masks parts of the sequence, and the model attempts to rebuild the missing segments from surrounding context. The score is the MSE in the masked region. The intuition is that normal context predicts normal signal more reliably than fault-related deviations, which can amplify anomaly contrast [28, 6].

6.5 Aggregation views and causal post-processing

Given per-timestep scores $s \in \mathbb{R}^{N \times L}$ for N windows, the evaluation reports several views:

- **Timestep view:** scores are compared directly against the binary mask m.
- **Window-max view:** a causal trailing max filter of width W is applied,

  $$\tilde{s}(t) = \max\{s(t - W + 1), \ldots, s(t)\},$$

  to reduce spurious short spikes without leaking future information.
- **Sequence-max view:** a window-level score is formed by taking $\max_t s(t)$ and the label is $\max_t m(t)$.

6.6 Threshold selection and detection latency

To turn scores into binary decisions (for reporting F1, latency, and event-style alerting metrics), the evaluation uses a fixed threshold selected without access to test labels. Concretely, a threshold is calibrated on a held-out subset of the training split (preferably by file), by taking a high quantile of the **sequence-max** score distribution on presumed-normal windows. This yields an operating point that corresponds to a low false-alarm rate under the normal regime and avoids test-set tuning.

False-alarm budget as a design choice. This report makes the false-alarm budget explicit by treating `evaluation.threshold_calibration_quantile` as an operational knob: choosing a higher quantile corresponds to targeting fewer false alarms on normal data. The alerting policy is then completed by `evaluation.alert_persistence` and `evaluation.alert_cooldown`, which implement a causal persistence/cooldown mechanism to suppress isolated score spikes and prevent repeated alerts. The precise budget (e.g., "≤ 1 false alert/day on normal-only streams") is a design choice that may be revisited depending on deployment requirements and the cost of investigation.

Quantiles to expected false alerts/day. If windows were independent and a threshold is set to the $(1 - \alpha)$ quantile of a normal-only score distribution, then the expected fraction of (normal) windows above threshold is α, where α is the per-window exceedance probability implied by the chosen quantile. For a window stride of S samples at sampling rate f_s (Hz), this corresponds to approximately

$$\text{windows/day} \approx \frac{86400\, f_s}{S} \quad \Rightarrow \quad \mathbb{E}[\text{false windows/day}] \approx \alpha\, \frac{86400\, f_s}{S}.$$

The implemented alert logic additionally requires *persistence*: an alert triggers only after p consecutive windows exceed threshold (where p is set by `evaluation.alert_persistence`,

converted internally to a number of window steps), and `evaluation.alert_cooldown` can suppress further alerts for a fixed time after triggering. Under a crude independence approximation, persistence reduces the expected alert rate roughly as

$$\mathbb{E}[\text{false alerts/day}] \approx \alpha^p \, \frac{86400 \, f_s}{S},$$

with cooldown further reducing the realised alert count when alerts are not rare.

For example, with $f_s = 28{,}000$ Hz and non-overlapping 3-cycle windows ($S = 1680$ samples, ≈ 0.06 s stride), there are $\approx 1.44 \times 10^6$ windows/day; a $(1 - \alpha) = 0.995$ calibration implies $\alpha = 0.005$, i.e., $\approx 7.2 \times 10^3$ *false windows/day* when $p = 1$, but only ≈ 0.18 *false alerts/day* when $p = 3$. In practice, scores are temporally dependent (and faults/load switching are structured), so the mapping from $(\alpha, p, \text{cooldown})$ to alerts/day is best treated as an empirical calibration step on long normal streams.

For comparison only, the runner can also report an **oracle** threshold that maximises F1 on the evaluated split. This provides an optimistic upper bound on thresholded metrics and is treated as such in the discussion.

Using the chosen threshold, detection latency is reported for fault-containing windows as the number of timesteps between the first true fault sample and the first threshold crossing.

This evaluation choice explicitly connects offline score separability to an operational objective: early threshold crossing with a low false-alarm rate.

Chapter 7

Experimental design

The protocol for training and evaluating diffusion models in this report is based on the configuration-driven implementation in `scripts/train_eval.py`. The aim is reproducible model comparison with metrics that map directly to implemented scoring and evaluation routines.

7.1 Configuration-driven experiments

All experiments are defined by YAML configuration files (e.g., `configs/lab_data/*.yaml`). Each run:

- Creates a timestamped output folder under `runs/`.
- Saves the resolved `config.yaml` used for that run.
- Saves the trained model parameters (`model.pt`).
- Writes a `metrics.json` summary and diagnostic plots.

The implementation is designed to be largely stateless and streaming: datasets are iterated without fully materialising them in RAM, allowing evaluation on very large test sets (e.g., millions of windows).

7.2 Data settings

Core data hyperparameters are:

- Window length L set by `data.seq_len`.
- Input channels C inferred from the dataset.
- Normalisation strategy and parameters set by `data.normalization_strategy`.

The implementation includes an explicit check that the configured `model.in_channels` matches the dataset output.

7.3 Model variants

The codebase supports both continuous (Gaussian) and discrete diffusion families. This report focuses on discrete diffusion with a token denoiser `TokenEmbeddingDenoiser` trained to predict clean tokens (categorical diffusion / D3PM-style) [3]. The token vocabulary size K and tokenisation layout (flattened vs unflattened) are specified in `diffusion.*`. Gaussian diffusion is implemented via `Denoiser1D` but is not the primary focus of these experiments.

7.4 Training protocol

Training uses the shared loop in `train_diffusion`:

- Mini-batch optimisation with Adam.
- Random diffusion timestep $t \sim \text{Uniform}(\{0, \ldots, T-1\})$ per example.
- Loss is cross-entropy for discrete diffusion.
- Cosine annealing learning-rate schedule across epochs.
- Gradient clipping and exponential moving average (EMA) of model parameters.

7.5 Evaluation protocol

Evaluation reports both *fidelity* (signal reconstruction quality) and *detection* (anomaly separability):

- A reconstruction timestep t_{rec} is chosen as a fraction of T via `evaluation.reconstruction_t_idx_fractior`
- Reconstruction and likelihood-based anomaly scores are computed per timestep.
- **Channel Aggregation**: Anomaly scores are aggregated across channels using the **max** function ($s = \max_c s_c$), which preserves sensitivity to single-phase faults better than averaging.
- Diagnostic plots include training loss curves, generated samples, score distributions, calibration curves, and example anomaly detections.

7.5.1 Baselines and comparisons

The primary baseline within this report is the discrete diffusion model with reconstruction- and likelihood-style scoring under a fixed evaluation protocol (windowing, calibration, and aggregation). Model comparisons therefore focus on controlled ablations of quantisation, embedding size, window length, and score family rather than on re-implementing classical detectors. For the lab dataset, external baselines are discussed qualitatively in Chapter 2; direct head-to-head baseline re-runs are out of scope for the current rerun. For benchmark datasets (MSL/SMAP/PSM), the reported scores follow the Anomaly-Transformer-style protocol to make them directly comparable to published baselines, even though those baselines are not re-trained here. We therefore focus on internal ablations and operational calibration/policy, and treat absolute performance claims as tentative. As immediate next work, we will add at least one ultra-cheap baseline computed from existing windows (e.g., a simple statistical thresholding rule) to anchor absolute performance. Gaussian-diffusion results in this report belong to this benchmark-comparability track and are not used as the primary lab-dataset modelling baseline.

Formal view of the evaluation pipeline. Let a multivariate signal be $x \in \mathbb{R}^{C \times T_{\text{raw}}}$. With window length L and stride S, the n-th extracted window is

$$x^{(n)} = x_{:,\, nS:nS+L-1} \in \mathbb{R}^{C \times L}.$$

For reconstruction-style evaluation, a fixed corruption level is chosen by

$$t_{\text{rec}} = \lfloor \rho\, T \rfloor, \qquad \rho = \texttt{evaluation.reconstruction_t_idx_fraction}.$$

Let $s_c^{(n)}(\ell)$ denote a per-channel per-timestep anomaly score (reconstruction residual, denoising CE, etc.) for window n, channel c, and local timestep $\ell \in \{0, \ldots, L-1\}$. The default lab-data aggregation is

$$s^{(n)}(\ell) = \max_{c \in \{1,\ldots,C\}} s_c^{(n)}(\ell).$$

For the sequence-max view used in threshold calibration and reporting, the window-level score is

$$S^{(n)} = \max_{\ell} s^{(n)}(\ell).$$

Threshold calibration (quantile protocol). On a held-out set of presumed-normal windows $\mathcal{D}_{\text{cal}}$, the calibrated-quantile threshold is

$$\tau = \text{Quantile}_q\Big(\{S^{(n)} : x^{(n)} \in \mathcal{D}_{\text{cal}}\}\Big), \qquad q = \texttt{evaluation.threshold_calibration_quantile}.$$

This is the default `calibrated_quantile` strategy and produces a fixed operating point without using test labels. For comparison only, the runner can also report `oracle_f1`, which selects the threshold that maximises F1 on the evaluated split; this is treated strictly as an optimistic upper bound.

7.5.2 Automated model selection and score-family choice

To keep the results reproducible and to avoid ad hoc checkpoint selection, the final protocol includes an explicit **model-selection sweep** and an **automated checkpoint selector**. The selection sweep varies a compact grid of core hyperparameters (bins K, token embedding size, quantiser strategy) with two repeated runs per setting under the same categorical cross-entropy objective, and computes metrics for all implemented score families used in the rerun (reconstruction and likelihood).

The selector then chooses a single *best* checkpoint using a normal-only objective on the *sequence-max view*: **minimise false alerts per day** on normal-only windows (`*.views.sequence_max.detectio`
If multiple score families are candidates, the selected pair (score family, checkpoint) is the one that yields the minimum normal-only FAR/day.

Checkpoint-selection pseudocode (high level).

```
for each checkpoint c and score family f in {recon, likelihood}:
  evaluate f on normal-only windows
  calibrate threshold tau(c, f) on a normal-only training holdout (quantile protocol)
  compute FAR_day(c, f) on normal-only eval windows using tau(c, f)
select the (c, f) pair with minimum FAR_day
```

Because these metrics are computed for each score family, the selection is also implicitly a *score-family choice*: the best model can be selected on reconstruction or likelihood scores depending on which achieves the best operational trade-off.

To keep selection tractable, this sweep uses a small test fraction (e.g., 5%) while maintaining the non-oracle calibration protocol. The selected checkpoint is then re-evaluated on a full or near-full test fraction in the dedicated evaluation sweeps.

7.5.3 Run orchestration and robustness

All experiments are launched via a suite manifest (`configs/experiment_suites/rerun_all.yaml`) and executed by `scripts/run_experiment_suite.py` (invoked as `python scripts/run_experiment_suite.py --suite configs/experiment_suites/rerun_all.yaml`). This provides:

- A single command to reproduce the full run list (ablations, benchmarks, evaluations).
- Automatic checkpoint selection for downstream eval-only runs.
- Robustness to partial failures (e.g., OOMs) by continuing to the next step.
- Aggregated logs and a summary of completed steps.

7.5.4 Compute budget and reproducibility

The experiments are constrained by a fixed compute budget, which motivates compact sweep grids and limited-epoch runs for model selection, followed by longer evaluations on the chosen checkpoints. Each run stores the resolved `config.yaml`, the model checkpoint, and a `metrics.json` summary in a timestamped folder under `runs/`, so results can be traced back to exact hyperparameters. Where supported in the configuration, runs fix random seeds and deterministic settings to reduce variance across reruns, with any deviations (e.g., mixed-precision or accelerated backends) documented in the run configs.

7.5.5 Fidelity-only mode

Some sweeps use a fidelity-only setting (`evaluation.skip_test_dataset: true`) where the fault test split is not loaded. To keep fidelity evaluation fair, a deterministic holdout subset is created from the training dataset, preferably by holding out entire files (`evaluation.fidelity_holdout_by_file`).

7.6 Evaluation metrics

Evaluation principles and metric rationale are discussed in Chapter 2; this chapter records the *implemented* protocol used for comparisons.

7.6.1 Fidelity diagnostics

Fidelity is evaluated on held-out presumed-normal windows (or the fidelity holdout in fidelity-only mode) using:

- time-domain metrics (MAE/MSE/RMSE and related summaries),
- frequency-domain metrics (spectral errors and convergence summaries),
- distributional summaries used as calibration/drift diagnostics.

7.6.2 Anomaly-detection metrics

Detection evaluation uses the score families described in Chapter 6 and reports:

- threshold-free ranking metrics (AUROC, PR-AUC) on timestep/window views;
- thresholded metrics under the non-oracle calibrated-quantile protocol (F1, detected-event fraction, latency, false-alert metrics);
- oracle-F1 thresholds only as optimistic diagnostics.

7.6.3 Operating-point protocol

For each run, a sequence-level score is calibrated on held-out presumed-normal windows using

$$\tau = \text{Quantile}_q\Big(\{S^{(n)} : x^{(n)} \in \mathcal{D}_{\text{cal}}\}\Big), \qquad q = \texttt{evaluation.threshold_calibration_quantile}. \tag{7.1}$$

False alerts per day are reported in operational units:

$$\text{FAR/day} = \text{FP_count} \times \frac{86400}{\text{total_normal_seconds}}. \tag{7.2}$$

This keeps model selection and reporting aligned with deployment constraints rather than test-label tuning.

7.6.4 Validity controls

The protocol applies split and contamination controls documented earlier in this chapter (file-aware holdouts, non-oracle calibration, and deterministic evaluation fractions where needed for compute). Together these controls aim to preserve comparability across sweeps while reducing optimistic bias from leakage or threshold overfitting.

Chapter 8

Results and evaluation

This chapter reports results from the experiment runner (`scripts/train_eval.py`) under the protocol in Chapter 7, covering the completed model-selection ablation on the lab dataset and benchmark evaluations used to position this project against common time-series anomaly-detection protocols. Unless stated otherwise, evaluations use deterministic test subsampling for throughput, with a dedicated full-test follow-up on 2026-02-06 included where needed. When reporting *false alerts per day*, rates are computed using Equation 7.2, where `total_normal_seconds` is the normal-only evaluation duration recorded in each run's metrics. Reported numbers use the final code path with max-channel score aggregation and corrected label-boundary handling.

Summary of findings. The completed model-selection sweep evaluates a 36-run grid over discrete quantisation and model-capacity settings on the lab dataset. Sequence-level threshold-free metrics show modest gains over chance in some settings, with the best likelihood-scoring configuration achieving AUROC 0.548 and PR-AUC 0.336 (baseline PR-AUC equals the label prevalence, 0.279). The same configuration is also competitive for reconstruction-based scoring (AUROC 0.515, PR-AUC 0.347). Across the grid, `cdf` (quantile) binning tends to improve likelihood-style scoring, while uniform binning can improve reconstruction PR-AUC but not necessarily AUROC. Overall separability remains limited, and calibrated operating points still imply unrealistically high alert rates on short evaluation spans, motivating follow-up work on more stable scoring and stronger discriminative signals. A full-test follow-up run on 2026-02-06 (Table 8.3) confirms the same near-random separability.

8.1 Model selection sweep (lab dataset)

The completed model-selection ablation is reported in `server_runs/ablation_model_selection_20260203_212612`. The sweep evaluates a $3 \times 3 \times 2 \times 2$ grid over:

- Quantisation bins $K \in \{128, 256, 512\}$.
- Token embedding dimension $d_{\text{embed}} \in \{32, 64, 128\}$.
- Quantiser strategy `uniform` vs `cdf` (CDF/quantile bins fit on training data only).
- Repeat index $r \in \{1, 2\}$: two repeated runs for each $(K, d_{\text{embed}}, \text{quantiser})$ setting under the same categorical cross-entropy training objective.

All runs use the lab dataset with $L = 1680$, $T = 400$, and $t_{\text{rec}} = 200$. For throughput, the sweep evaluates a deterministic 0.5% test subsample with `evaluation.likelihood_n_`

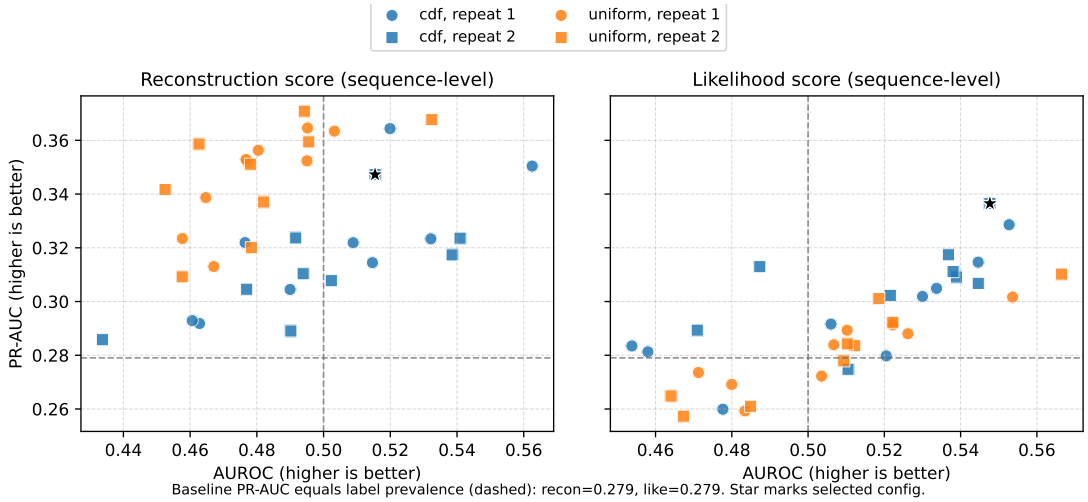


Figure 8.1: Model-selection sweep (lab dataset): sequence-level threshold-free metrics (AUROC vs PR-AUC) for reconstruction and likelihood score families. Each point is one configuration; colours/markers encode quantiser strategy and repeated-run index. The dashed PR-AUC baseline equals the label prevalence. The star marks the selected configuration (best likelihood PR-AUC).

`samples=1`. Thresholded summaries use normal-only calibrated quantiles ($q = 0.995$, file-wise split).

8.1.1 Score families, views, and selection criteria

The sweep reports reconstruction and likelihood-style scores over timestep, causal-window, and sequence-max views. Selection emphasises sequence-level AUROC/PR-AUC (with prevalence as PR baseline), while thresholded metrics are reported under the non-oracle calibrated-quantile protocol [13, 39]. Gradient and inpainting scores are implemented but not included in this sweep.

8.1.2 Threshold-free results across the grid

Figure 8.1 plots sequence-level AUROC vs PR-AUC for every configuration under reconstruction and likelihood scoring. The dashed horizontal line shows the label prevalence (the random baseline for PR-AUC). Many settings cluster close to AUROC 0.5 and PR-AUC near prevalence, but a subset achieve consistent improvements.

To expose the main effects of quantisation resolution K and embedding capacity, Figure 8.2 averages metrics over the other factors and shows mean AUROC/PR-AUC as a function of (K, d_{embed}).

8.1.3 What worked (and why)

Table 8.1 summarises mean metrics by factor level. Two patterns stand out:

- **CDF binning helps likelihood-style scoring.** The likelihood score is an average cross-entropy over tokens; CDF/quantile bins approximately equalise token frequencies, reducing rare-token dominance and making the likelihood proxy more informative under imbalance. This improves likelihood AUROC and PR-AUC on average.

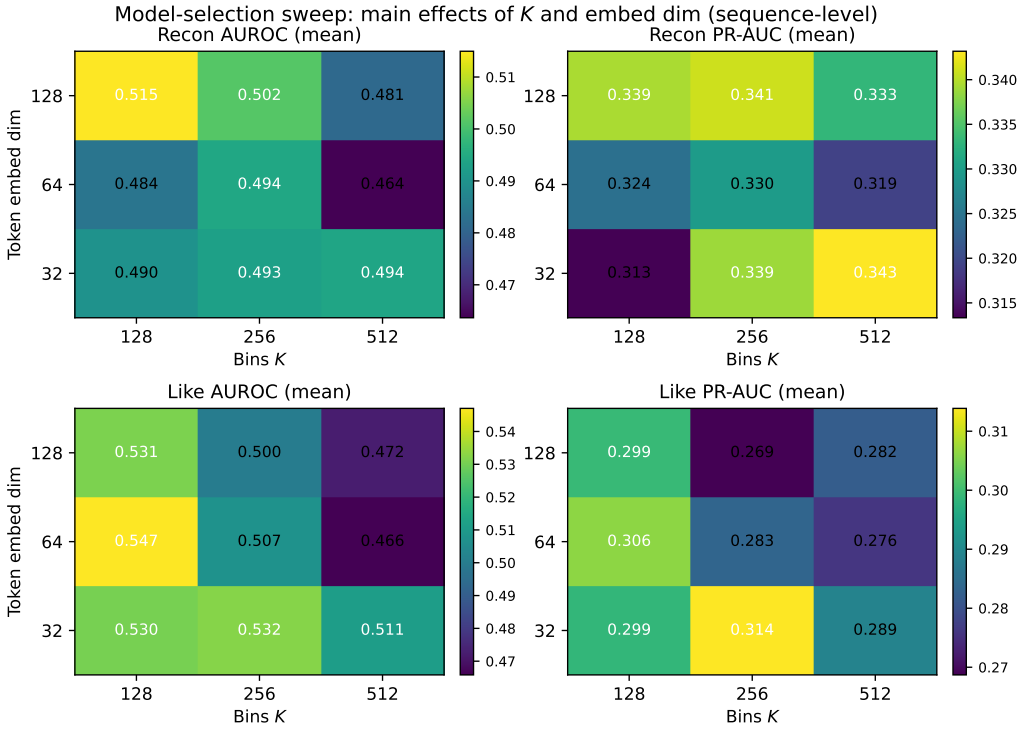


Figure 8.2: Model-selection sweep (lab dataset): mean sequence-level AUROC and PR-AUC as a function of quantisation bins K and token embedding dimension d_{embed}, averaged over quantiser strategy and repeated-run index.

- **Uniform binning can help reconstruction PR-AUC.** Reconstruction scores are computed as squared error in the *dequantised amplitude domain*. Uniform bins preserve a roughly linear amplitude scale, so squared-error residuals behave more like a physically meaningful distance. With CDF bins, the dequantisation map is non-linear, which can compress large-amplitude regions and distort the residual scale, plausibly reducing reconstruction PR-AUC even when AUROC is competitive [18, 51].

The repeated-run factor has only a small mean effect, indicating limited run-to-run variance under this setup.

Table 8.1: Model-selection sweep: mean sequence-level AUROC/PR-AUC by factor level (averaged across the full grid).

Factor level	Recon AUROC	Recon PR-AUC	Like AUROC	Like PR-AUC
Quantiser strategy				
`cdf`	0.501	0.316	0.515	0.300
`uniform`	0.481	0.346	0.506	0.281
Repeat run				
`repeat_1`	0.491	0.333	0.507	0.287
`repeat_2`	0.490	0.329	0.514	0.294

8.1.4 Selected configuration

Table 8.2 lists the best configurations under each score family and criterion. The selected setting is $K = 256$, $d_{\text{embed}} = 32$ with `cdf` binning, which is the best likelihood PR-AUC

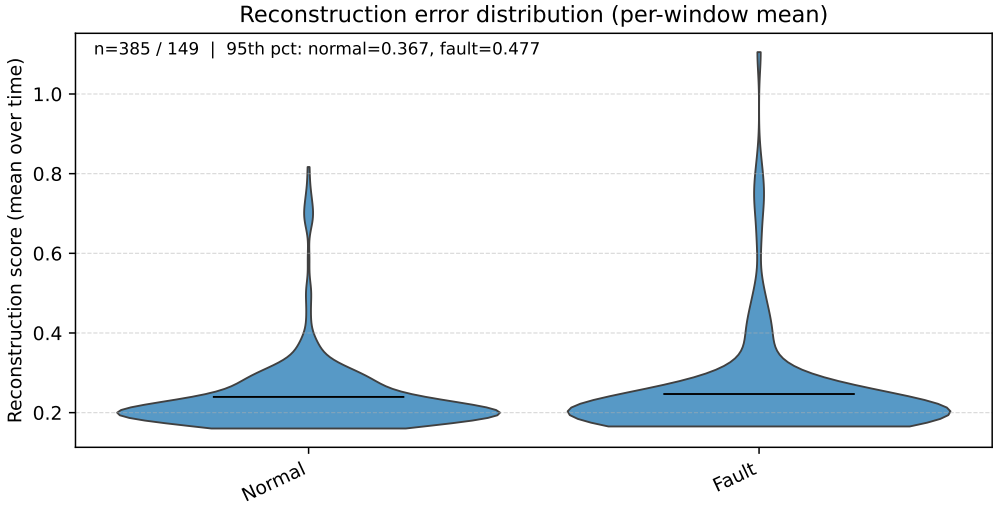


Figure 8.3: Distributional diagnostic: reconstruction-error distribution (sequence-level mean score) for normal vs fault windows. A strong detector shows clear separation (fault scores shifted higher with limited overlap).

configuration (AUROC 0.548, PR-AUC 0.336) and remains competitive for reconstruction scoring (AUROC 0.515, PR-AUC 0.347). This choice is also computationally attractive (small embedding dimension) and avoids settings that achieve high PR-AUC without improving AUROC.

Table 8.2: Model-selection sweep: best configurations by score family and criterion (sequence-level view).

Family	Criterion	K	Embed	Quantiser	Repeat	AUROC ↑	PR-AUC ↑
Reconstruction	max PR-AUC	512	32	`uniform`	`repeat_2`	0.494	0.371
Reconstruction	max AUROC	128	128	`cdf`	`repeat_1`	0.563	0.350
Likelihood	max PR-AUC	256	32	`cdf`	`repeat_2`	0.548	0.336
Likelihood	max AUROC	128	64	`uniform`	`repeat_2`	0.566	0.310
Selected (likelihood)	selected	256	32	`cdf`	`repeat_2`	0.548	0.336

8.1.5 Qualitative diagnostics and failure cases

The runner exports reconstruction examples, class-conditional score distributions, calibration plots, and timeline views. These diagnostics are used to distinguish threshold issues from intrinsic score-overlap/saturation failures.

Distributional diagnostics (violin + snapshots). For distribution-level error analysis, the runner can export per-window score distributions and a small set of representative reconstructions (normal vs fault, each with high/low reconstruction error). When available, the resulting figures provide (i) violin plots comparing the reconstruction-error distributions for normal vs fault windows and for fault subtypes, and (ii) signal snapshots showing true vs reconstructed signals, absolute error, and per-timestep scores with the true fault mask overlaid.

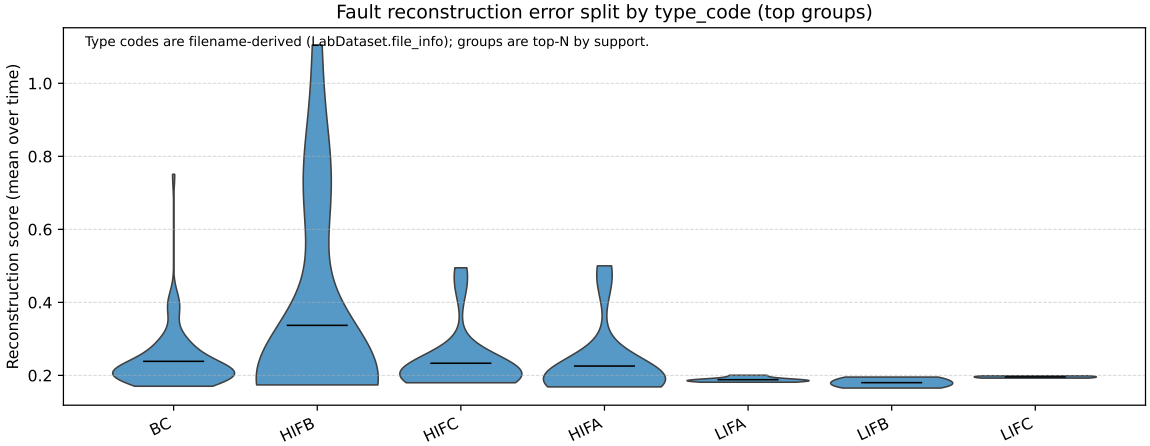


Figure 8.4: Distributional diagnostic: reconstruction-error distributions for fault windows split by fault type code (top groups by support). This highlights fault classes that are consistently more or less separable under residual scoring.

8.1.6 Operational and sensitivity ablations (summary)

Detailed ablation tables/figures are moved to Appendix A (Sections A.1–A.2) so this chapter can focus on main outcomes.

Key outcomes from those sweeps are:

- **Threshold quantile sweep:** moving q from 0.99 to 0.997 changed the calibrated threshold but left sequence-level operating metrics essentially unchanged (F1 ≈ 0.409, FAR/day(normal) $\approx 2.5 \times 10^4$, detected fraction ≈ 0.062; Table A.1, Figure A.1).

- **Alert policy sweep:** stronger persistence reduced FAR/day by about $13.9\times$ ($\approx 2.5 \times 10^4 \rightarrow 1.8 \times 10^3$) but also reduced detected-event fraction ($0.062 \rightarrow 0.021$); cooldown changes had negligible effect in this run (Table A.2, Figure A.2).

- **Initial-noise sweep:** increasing reconstruction corruption ($\rho : 0.5 \rightarrow 0.9$) improved reconstruction PR-AUC ($0.300 \rightarrow 0.544$) but AUROC remained near random (0.487–0.501), indicating persistent overlap/saturation (Figure A.3, Figure A.4, Table A.3).

- **Window-size-oracle rerun:** no valid metrics were produced because the planned run referenced an unresolved checkpoint path (`server_runs/ablation_window_sizes_oracle_eval_20260205_122220`).

Full-test selected-checkpoint evaluation (2026-02-06). To replace the missing full-checkpoint rerun stage, a dedicated one-run full-test evaluation was executed in `server_runs/ablation_submission_best_checkpoint_full_eval_20260206_130720`. This run evaluates a fixed checkpoint used in the submission follow-up ($K = 512$, token embed $= 128$, `cdf` binning, cross-entropy objective) over the full labelled test split (106,782 windows; 573 labelled fault events). Table 8.3 shows that threshold-free separability is close to random for both score families. Operationally, reconstruction scoring at the calibrated threshold missed all labelled events (0/573) while still producing FAR/day(normal) $\approx 3.06 \times 10^3$; likelihood scoring detected 23/573 events (detected fraction 0.040) but with FAR/day(normal) $\approx 7.29 \times 10^4$. These full-test results reinforce that the core bottleneck is score separability, not just threshold selection.

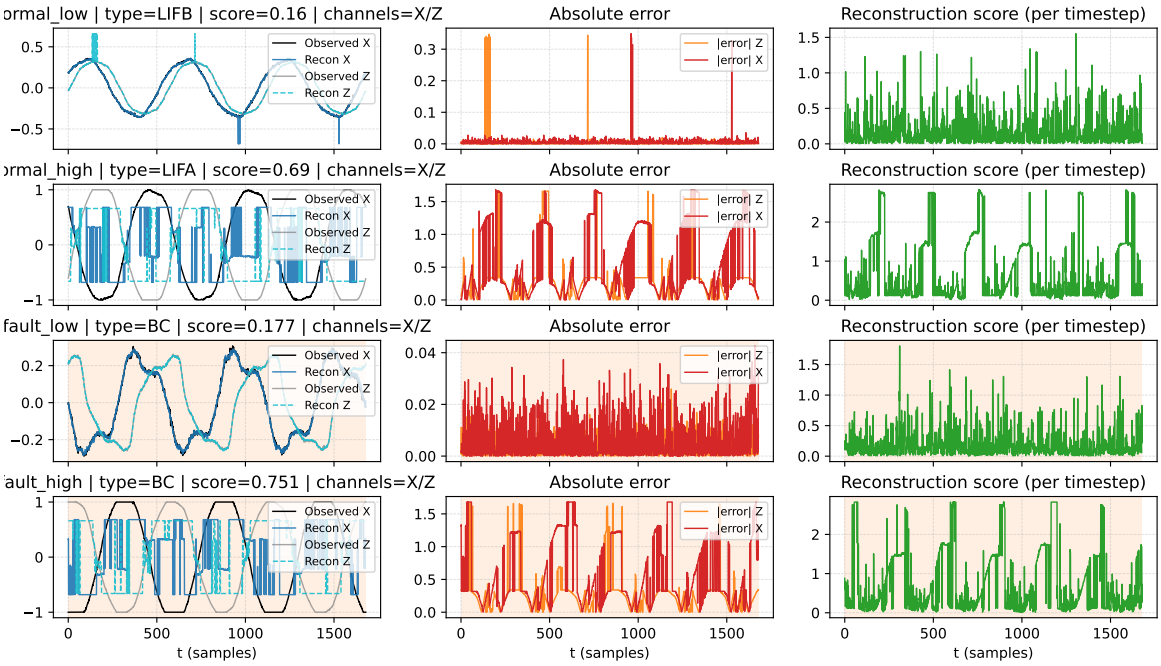


Figure 8.5: Qualitative snapshots: true vs reconstructed raw sensor channels (X and Z), absolute error, and per-timestep reconstruction score for representative normal and fault windows with high/low error. Shaded regions indicate labelled fault spans.

Table 8.3: Full-test follow-up (2026-02-06): sequence-level metrics for the submission follow-up checkpoint on the full labelled test split.

Score family	AUROC ↑	PR-AUC ↑	F1 ↑	FAR/day (normal) ↓	Detected events	Detected fraction
Reconstruction	0.493	0.277	0.413	3063.6	0/573	0.000
Likelihood	0.498	0.267	0.408	72906.3	23/573	0.040

8.1.7 External baseline: supervised fault classifier (prior work)

For context, a prior student project trained a *supervised* deep neural network (DNN) to classify the same lab dataset into 15 classes (one `No-Fault` class plus 14 fault categories). Their results are summarised in the provided document `cfat_DNN_confusion-matrix.docx`, which contains a classification report (precision/recall/F1 with support) and a row-normalised confusion matrix.

This baseline is not directly comparable to the one-class diffusion setting in this report: it uses fault labels during training, optimises an explicit discriminative objective, and predicts fault *type* rather than detecting faults under label-sparse constraints. It is therefore best interpreted as an **upper bound** on what is achievable when fault-labelled training data and a supervised objective are available.

To align with the anomaly-detection framing, we collapse the multiclass classifier into a **binary** decision: "fault" (any non-`No-Fault` class) vs "no-fault". Using the reported confusion matrix and per-class supports, the resulting binary metrics are shown in Table 8.4. Under this collapse, the supervised baseline achieves accuracy 0.941 and F1 0.941, with fault recall 0.930 and specificity 0.953.

Figure 8.7 shows which fault categories are most often predicted as `No-Fault` (binary false negatives). These classes represent the hardest faults for the supervised classifier and

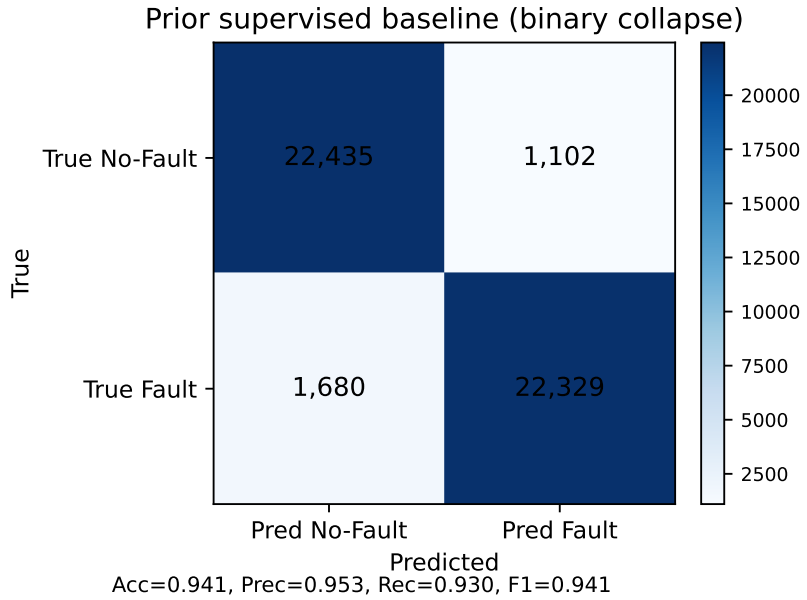


Figure 8.6: Binary confusion matrix for the prior supervised DNN baseline after collapsing all fault categories into a single "fault" class. Counts are derived from the row-normalised confusion matrix and per-class supports in `cfat_DNN_confusion-matrix.docx`.

are plausible candidates for "low-contrast" anomalies that also challenge generative residual scoring.

Table 8.4: Prior supervised DNN baseline on the lab dataset. Multiclass accuracy is reported in the original document. Binary metrics are computed by collapsing all fault classes into a single "fault" label using the confusion matrix first column (predicted `No-Fault`) and the per-class supports.

Metric	Value	TN	FP	FN
Multiclass accuracy	0.900	–	–	–
Binary (fault vs no-fault) accuracy	0.941	22,435	1,102	1,680
Binary precision (fault)	0.953	–	–	–
Binary recall (fault)	0.930	–	–	–
Binary F1 (fault)	0.941	–	–	–
Binary specificity (no-fault)	0.953	–	–	–

8.2 Benchmark datasets (MSL/SMAP/PSM/SMD)

To connect these results to common time-series anomaly-detection benchmarks, evaluations were run on MSL, SMAP, PSM, and SMD under the Anomaly-Transformer-style protocol (stride-1 windows, last-timestep alignment, and point-adjusted metrics) [56]. Two diffusion families are reported: **Gaussian diffusion** (the Gaussian benchmark runs) and **discrete diffusion**. All benchmark numbers are taken from the `aligned_last` view. Point-adjusted (PA) metrics follow the standard benchmark convention: if any point in a labelled anomaly segment is detected, the entire segment is counted as detected, which can substantially inflate segment recall relative to raw pointwise precision/recall/F1. These benchmark runs

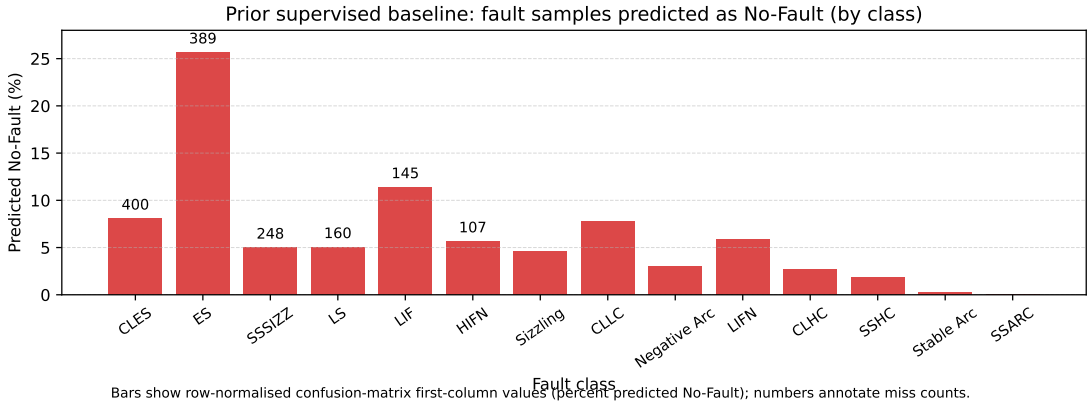


Figure 8.7: Prior supervised DNN baseline: fraction of each fault class predicted as `No-Fault` (binary false negatives), annotated with the corresponding miss counts.

are used as protocol-comparability exercises with prior time-series AD literature, not as evidence that the approach already solves deployment-grade HIF detection on the lab dataset.

Figure 8.8 shows the main pattern in this setup: discrete diffusion often achieves high PA F1 even when pointwise F1 remains low, indicating "segment hits" without precise localisation. Gaussian diffusion is generally more consistent between pointwise and PA metrics, and achieves higher AUROC/PR-AUC on most datasets in this snapshot.

Table 8.5: Benchmark results (Gaussian diffusion): reconstruction-score metrics under the benchmark protocol. Precision/recall/F1 are raw pointwise values and point-adjusted (PA) values.

Dataset	AUROC ↑	PR-AUC ↑	Prec ↑	Recall ↑	F1 ↑	Prec (PA) ↑	Recall (PA) ↑	F1 (PA) ↑
MSL	0.622	0.188	0.154	0.497	0.235	0.268	1.000	0.423
SMAP	0.567	0.160	0.152	0.681	0.249	0.208	1.000	0.345
PSM	0.765	0.529	0.467	0.774	0.583	0.531	0.999	0.694
SMD	0.733	0.183	0.342	0.223	0.270	0.682	0.921	0.784

Table 8.6: Benchmark results (Gaussian diffusion): likelihood-score metrics under the benchmark protocol. Precision/recall/F1 are raw pointwise values and point-adjusted (PA) values.

Dataset	AUROC ↑	PR-AUC ↑	Prec ↑	Recall ↑	F1 ↑	Prec (PA) ↑	Recall (PA) ↑	F1 (PA) ↑
MSL	0.621	0.183	0.151	0.520	0.234	0.255	1.000	0.407
SMAP	0.573	0.164	0.152	0.822	0.257	0.179	1.000	0.304
PSM	0.752	0.504	0.458	0.763	0.572	0.525	0.999	0.688
SMD	0.727	0.185	0.288	0.219	0.248	0.616	0.867	0.720

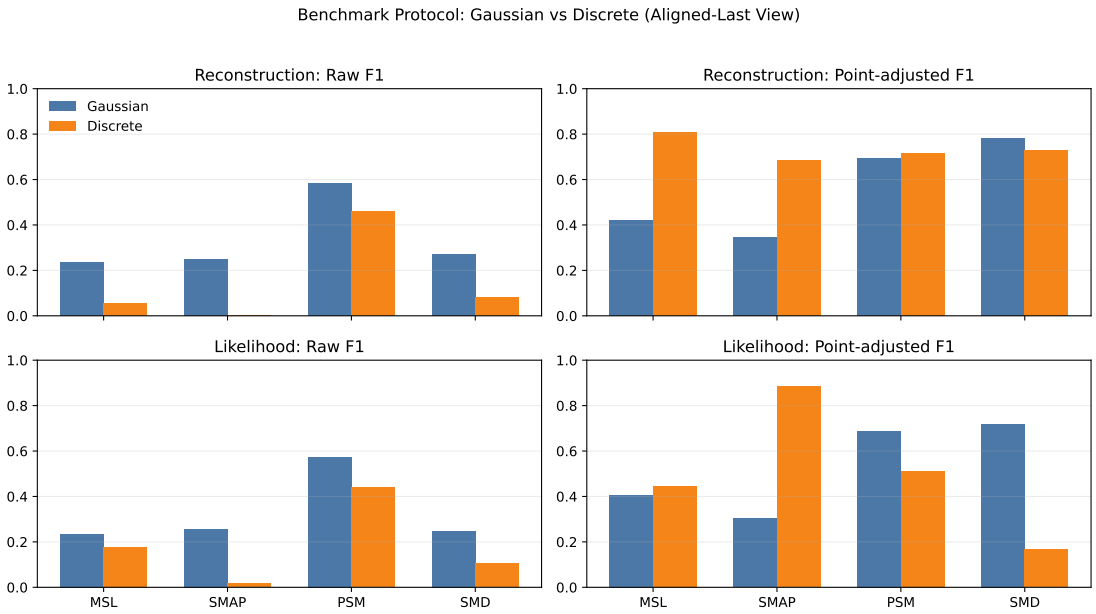


Figure 8.8: Benchmark protocol comparison (aligned-last): Gaussian vs discrete diffusion for reconstruction and likelihood scores, showing raw pointwise F1 vs point-adjusted (PA) F1. High PA F1 with near-zero raw F1 indicates that the detector often triggers somewhere within anomaly segments but does not localise anomalies well at the point level.

Table 8.7: Benchmark results (discrete diffusion): reconstruction-score metrics under the benchmark protocol. Precision/recall/F1 are raw pointwise values and point-adjusted (PA) values.

Dataset	AUROC ↑	PR-AUC ↑	Prec ↑	Recall ↑	F1 ↑	Prec (PA) ↑	Recall (PA) ↑	F1 (PA) ↑
MSL	0.529	0.140	0.398	0.031	0.057	0.939	0.710	0.808
SMAP	0.507	0.129	0.095	0.002	0.003	0.970	0.531	0.687
PSM	0.638	0.394	0.409	0.521	0.458	0.565	0.978	0.716
SMD	0.607	0.101	0.183	0.052	0.081	0.754	0.705	0.729

Table 8.8: Benchmark results (discrete diffusion): likelihood-score metrics under the benchmark protocol. Precision/recall/F1 are raw pointwise values and point-adjusted (PA) values.

Dataset	AUROC ↑	PR-AUC ↑	Prec ↑	Recall ↑	F1 ↑	Prec (PA) ↑	Recall (PA) ↑	F1 (PA) ↑
MSL	0.536	0.123	0.120	0.338	0.177	0.287	1.000	0.446
SMAP	0.476	0.119	0.130	0.009	0.018	0.930	0.841	0.884
PSM	0.541	0.278	0.301	0.816	0.440	0.345	1.000	0.513
SMD	0.609	0.055	0.057	0.603	0.104	0.091	0.999	0.167

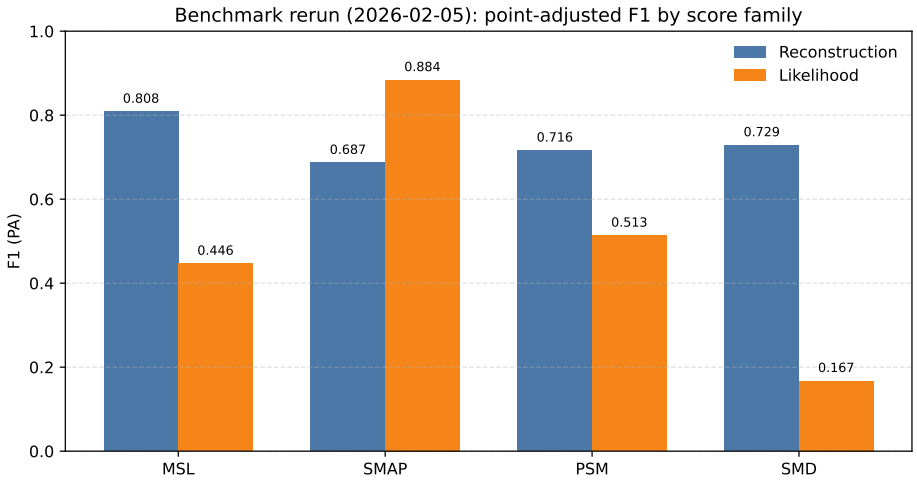


Figure 8.9: Benchmark rerun (2026-02-05, discrete diffusion): point-adjusted F1 (PA) by dataset and score family.

Table 8.9: Benchmark summary (aligned-last): point-adjusted F1 (PA) by score family.

Model / score	MSL	SMAP	PSM	SMD
Gaussian: Recon F1 (PA)	0.423	0.345	0.694	0.784
Gaussian: NLL F1 (PA)	0.407	0.304	0.688	0.720
Gaussian: Best (PA)	0.423	0.345	0.694	0.784
Discrete: Recon F1 (PA)	0.808	0.687	0.716	0.729
Discrete: NLL F1 (PA)	0.446	0.884	0.513	0.167
Discrete: Best (PA)	0.808	0.884	0.716	0.729

For reproducibility, the discrete rerun values are sourced from `server_runs/benchmark_overnight_20260205_191239/results.csv` (MSL/PSM/SMD) and `server_runs/rerun_all_20260205_112724/09_benchmark_smap.log` (SMAP).

Table 8.10: Benchmark rerun summary (2026-02-05, discrete diffusion): aligned-last metrics across MSL/SMAP/PSM/SMD.

Dataset	Recon AUROC	Recon PR-AUC	Recon F1	Recon F1 (PA)	NLL AUROC	NLL PR-AUC	NLL F1	NLL F1 (PA)
MSL	0.529	0.140	0.057	0.808	0.536	0.123	0.177	0.446
SMAP	0.507	0.129	0.003	0.687	0.476	0.119	0.018	0.884
PSM	0.638	0.394	0.458	0.716	0.541	0.278	0.440	0.513
SMD	0.607	0.101	0.081	0.729	0.609	0.055	0.104	0.167

Comparison to Anomaly Transformer (Xu et al., 2022). Xu et al. report adjusted F1 scores (as %) on the same four benchmarks and compare against ten competitive baselines in their main paper (including THOC and InterFusion) [56]. Their repository additionally reports results for a broader set of 15 baselines on these datasets [55]. Table 8.11 contrasts the best point-adjusted F1 achieved in this work (from Table 8.9) with the Anomaly Transformer F1 values from Table 1 of Xu et al. (2022) [56]. The comparison is indicative rather than strictly equivalent: differences in anomaly scoring and threshold selection can shift adjusted/PA metrics.

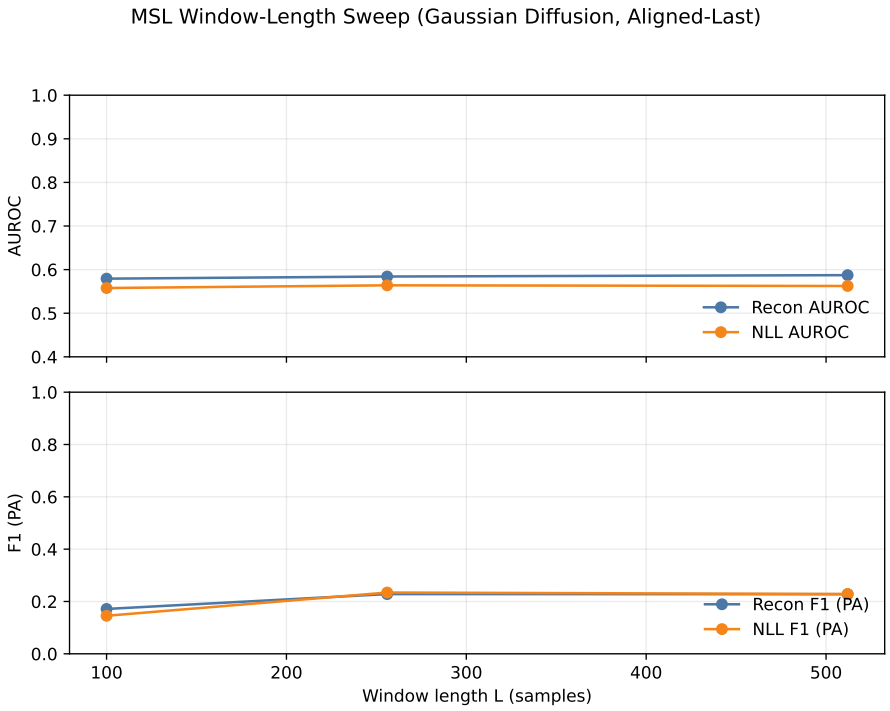


Figure 8.10: MSL window-length sweep (Gaussian diffusion, aligned-last): AUROC and point-adjusted F1 (PA) as a function of window length L.

Table 8.11: Benchmark comparison to Anomaly Transformer (Xu et al., 2022). This work reports best point-adjusted F1 (PA) from the aligned-last protocol; Anomaly Transformer values are adjusted F1 from the original paper.

Dataset	This work: Best PA F1 (%)	Anomaly Transformer F1 (%)	Gap (pp)
MSL	80.8	93.59	12.8
SMAP	88.4	96.69	8.3
PSM	71.6	97.89	26.3
SMD	78.4	92.33	13.9

Interpretation. Across these benchmarks, Gaussian diffusion provides the most consistent pointwise separability in this project snapshot (AUROC/PR-AUC are generally higher and raw F1 is non-trivial on PSM and SMD). Discrete diffusion is more brittle under the same protocol: it can yield high PA F1 (segment detection) while raw pointwise F1 stays low, notably on SMAP where AUROC is near random despite strong PA scores. This pattern suggests that, with the current thresholding and scoring, the discrete model often triggers somewhere within anomaly segments but does not reliably localise anomalous points.

Window-length sensitivity (MSL, Gaussian). To test whether the short benchmark window ($L = 100$) explains weaker benchmark scores, a sweep over $L \in \{100, 256, 512\}$ was run on MSL using the same benchmark protocol and the anomaly-ratio threshold heuristic. Table 8.12 reports aligned-last metrics. AUROC and PR-AUC remain nearly unchanged with larger windows, while point-adjusted F1 improves modestly from 0.171 at $L = 100$ to about 0.23 for $L \geq 256$ (Figure 8.10). This suggests that benchmark performance in the current setup is not primarily driven by window length alone.

Table 8.12: MSL benchmark window-length sweep (Gaussian diffusion, aligned-last protocol). Metrics shown for reconstruction and likelihood scores. Thresholds use the anomaly-ratio heuristic.

L	Recon AUROC	Recon PR-AUC	Recon F1	Recon F1 (PA)	NLL AUROC	NLL PR-AUC	NLL F1	NLL F1 (PA)
100	0.579	0.175	0.051	0.171	0.558	0.164	0.051	0.145
256	0.584	0.172	0.054	0.228	0.564	0.160	0.054	0.234
512	0.587	0.166	0.056	0.228	0.562	0.154	0.055	0.228

Chapter 9

Real-time performance and deployment considerations

Deployment of the diffusion-based detector in a streaming setting is discussed here, with emphasis on choices implemented in `scripts/train_eval.py` and `src/eds_diffusion/metrics.py`. The practical question is not whether diffusion can be trained, but whether anomaly scores can be produced at required throughput with thresholds that remain stable under drift.

9.1 Streaming interface and latency budget

The implemented system scores fixed-length windows of L samples, where L is configured by `data.seq_len`. As discussed in Chapter 2, the window length and stride jointly determine the context available to the model, the update rate of scores, and the earliest possible alert time. The evaluation code also supports the causal trailing max aggregation introduced in Chapter 6, which provides a simple persistence mechanism without requiring future context.

9.2 What makes inference expensive (and what does not)

Diffusion models are often associated with expensive multi-step sampling. In this project, anomaly scoring is intentionally designed to avoid full ancestral sampling in the deployment path. Where sampling is required (e.g., for qualitative diagnostics), step-reduced procedures such as DDIM-style updates or fast solvers can reduce the number of reverse steps [45, 32].

9.2.1 Single-pass reconstruction at a fixed timestep

The reconstruction score used for detection reconstructs $\hat{x}_0$ from a single forward corruption step at a configured timestep t_{rec} (typically $\lfloor 0.5T \rfloor$). This requires:

1. One forward corruption operation $q(x_t \mid x_0)$.

2. One denoiser forward pass to estimate clean tokens and dequantise $\hat{x}_0$.

This is far cheaper than running T reverse steps, and it scales linearly with the number of windows scored.

9.2.2 Likelihood-style scoring as Monte Carlo denoising error

In deployment, this suggests an adaptive policy: use the cheap reconstruction score continuously, and compute likelihood-style scores only when the reconstruction score is elevated (a two-stage detector), which helps constrain unnecessary alerts and operator load [19, 48].

9.2.3 Gradient and inpainting compute costs

The newly introduced scoring methods increase compute requirements:

- **Gradient scoring** requires three forward passes (original and two perturbed versions) to compute finite differences, increasing compute cost by ˜3× relative to reconstruction.
- **Inpainting scoring** requires a multi-step reverse sampling chain for each sliding window. Even with efficient batching, this scales with the number of sampling steps (e.g., 50–400) and is significantly more expensive than single-pass scores.

On edge devices like the Raspberry Pi 5, these high-fidelity scores should be reserved for refining "suspicious" windows flagged by the baseline reconstruction detector to avoid excessive alerting burden [19, 48].

9.3 Batching and hardware considerations

All scoring operations are vectorisable across windows. In a deployment pipeline that buffers a short batch of consecutive windows, the denoiser can be executed in batches to amortise overhead and saturate available compute (GPU or CPU). This is particularly important for the likelihood-style score, which repeats denoiser calls.

9.3.1 Raspberry Pi 5: CPU-first optimisation

For the target deployment platform considered in this report (Raspberry Pi 5), the practical constraint is that inference is typically executed on an ARM CPU without access to CUDA-style GPU acceleration. As a result, techniques that reduce parameter count but rely on specialised sparse GPU kernels (e.g., unstructured magnitude sparsity) are unlikely to translate into proportional latency improvements.

Two practical levers on this class of device are:

- **Quantisation:** integer (e.g., INT8) or reduced-precision inference can reduce memory bandwidth pressure on CPU-first deployments, provided the operator set is supported. In practice, the most robust deployment path is usually to export the model to an inference runtime with mature quantisation support (e.g., TensorFlow Lite or ONNX Runtime) and validate the end-to-end score stability under quantisation [18, 51].
- **Width reduction:** smaller denoisers usually give more predictable CPU latency gains than sparsity-oriented methods.

For diffusion-style scoring, the dominant deployment consideration remains the number of denoiser evaluations per window. On Raspberry Pi 5, this motivates (i) favouring single-pass reconstruction scores, (ii) keeping the Monte Carlo sample count `n_samples` small (or conditional on an elevated reconstruction score), and (iii) preferring smaller denoisers over marginal gains from sparsity that do not map cleanly to CPU speed.

The discrete diffusion variant introduces a token embedding and classification head, and operates on categorical tensors rather than floating point waveforms. While discrete inference avoids continuous noise prediction, it can increase memory bandwidth due to larger intermediate activations (embedding dimension D) and logits over K bins.

9.4 Causal post-processing and alerting

Raw per-timestep scores can be spiky. The evaluation code implements a causal trailing max filter of width W to reduce sensitivity to short-lived spikes while preserving real-time causality. This pattern is attractive for deployment because:

- It can be implemented online with constant memory.
- It naturally enforces persistence without introducing a separate rule engine.
- It makes detection latency measurable and tunable via W.

More conservative alerting can be achieved by adding a cooldown period after an alert (to avoid repeated alarms for the same event) and by requiring the filtered score to exceed threshold for multiple successive windows. These policies are best tuned against operational false-alarm budgets and alarm-fatigue constraints [19, 48].

9.5 Thresholding, calibration, and drift

Threshold maintenance is the main operational challenge for one-class detectors under non-stationarity [14, 50]. In the experimental code, thresholded metrics are reported using a non-oracle calibration protocol by default (training-holdout quantile), with an oracle F1-maximising threshold available only as an optimistic diagnostic upper bound. In deployment, thresholds should instead be set and recalibrated using presumed-normal data to target an acceptable false-alarm rate.

Two practical mechanisms are supported by the existing evaluation artifacts:

- **Score distribution monitoring:** track summary statistics of scores over time (mean/quantiles) under presumed-normal periods to detect drift [14, 50].
- **Calibration diagnostics:** the runner generates reliability diagrams by min–max scaling scores to $[0, 1]$ and computing calibration error metrics. While this is not a probability calibration in the strict sense, it provides a lightweight check for score stability.

When drift is detected, the least risky update is often recalibration (adjusting thresholds) rather than immediate model retraining [14, 50]. If retraining is performed, it should avoid contamination by recent fault periods; the project's fidelity-only evaluation mode includes a file-level holdout split precisely to reduce optimistic bias when labelled faults are unavailable.

9.6 Summary

The deployment path implemented in this report is designed to keep diffusion inference practical by using single-pass reconstructions and controlled Monte Carlo scoring rather than full T-step sampling. The remaining engineering work is therefore primarily about (i) choosing windowing and post-processing that meet latency constraints and (ii) maintaining thresholds under drift using normal-only monitoring and periodic recalibration.

Chapter 10

Discussion and limitations

The discussion below interprets the experimental results for diffusion-based anomaly detection in distribution-monitoring time series, with emphasis on which design choices helped, which did not, and what still blocks deployment readiness.

10.1 Implementation pipeline

The core pipeline can be summarised as:

1. Extract fixed-length windows from multi-channel magnetic-field recordings and normalise using statistics computed on training files.
2. Train a discrete diffusion denoiser (token prediction) on normal windows.
3. Score test windows using (i) single-pass reconstruction residuals at a fixed corruption level and (ii) a Monte Carlo denoising-loss proxy for likelihood.
4. Aggregate scores causally (optional window-max) and select thresholds for reporting.

In the deployed scoring path, we do not run full T-step sampling (Chapter 9). In practice, the scores mostly reflect how well the denoiser can map a partially corrupted window back toward the normal manifold.

10.2 Reconstruction scores versus likelihood-style scores

Reconstruction residuals are intuitive and cheap. In this project they are computed by corrupting to a fixed timestep t_{rec} and estimating $\hat{x}_0$ in a single model pass (Chapter 6), which makes them attractive for streaming deployments.

However, reconstruction scores inherit several known failure modes [54, 58]:

- **Over-general reconstruction:** a sufficiently expressive model may reconstruct anomalous windows well, compressing score separation.
- **Metric sensitivity:** pointwise MSE penalises phase shifts and alignment errors that may not be operationally anomalous. In this quasi-periodic AC setting, small window-alignment variability is plausible, so phase offsets can inflate residual scores without corresponding fault severity.
- **Choice of corruption level:** t_{rec} acts like a hyperparameter controlling how aggressively the model is forced to "repair" the signal.

This sensitivity to reconstruction corruption is visible in the initial-noise ablation (Appendix A.2, Figure A.3, Figure A.4, Table A.3): increasing ρ from 0.5 to 0.9 improves reconstruction PR-AUC (0.300 $\rightarrow$ 0.544), but AUROC remains near random (0.487–0.501) under persistent score overlap. In other words, changing corruption level changes score spread, but does not by itself solve normal–fault separability.

The likelihood/ELBO-proxy score addresses some of these issues by probing denoising difficulty across multiple timesteps rather than at a single fixed corruption level (Chapter 6). This tends to produce a score that is less tied to any one reconstruction and more tied to general model mismatch. The trade-off is compute: Monte Carlo averaging requires multiple denoiser passes.

However, ELBO/denoising-loss proxies still face the OOD likelihood paradox: generative models can assign higher model likelihood to out-of-distribution samples than to in-distribution data [34]. The discrete formulation does not remove this issue. In this project, forward-backward scores (forward corruption followed by backward reconstruction or inpainting) are therefore treated as the main deployment signals. They measure denoising difficulty rather than true likelihood and are empirically less vulnerable to that paradox.

In practice, the two score families play different roles: reconstruction provides a fast first-pass score, while likelihood-style scoring is a slower secondary check when a window already appears suspicious (Chapter 9).

10.3 Discrete diffusion in this project

The discrete formulation introduces an explicit quantisation step and turns denoising into token classification. The main advantages are:

- **Bounded representation:** inputs are explicitly confined to a finite vocabulary, which can reduce sensitivity to small analogue drift.
- **Interpretable likelihood proxy:** cross-entropy over tokens is a direct mismatch measure between the predicted clean sequence and the quantised observation.

The costs and risks are also clear in the implementation:

- **Quantisation as inductive bias:** too few bins can erase subtle early-stage fault signatures; too many bins increases classifier difficulty and memory/compute.
- **Uniform replacement corruption:** the current forward process uses random replacement, which may be suboptimal when token bins are ordinal (amplitude locality).

These points motivate a testable mechanism: if corruption respects ordinal bin locality, subtle departures should be harder to denoise than benign nearby amplitudes, improving early-fault separation. Future variants should therefore ablate locality-aware transition kernels (still within the D3PM framework) against uniform replacement and test whether normal–fault overlap is reduced in the target window/sequence view [3, 18, 51].

Model selection insights. The completed model-selection sweep shows that score-family choice interacts with quantisation. CDF/quantile binning tends to improve likelihood-style scoring, consistent with more balanced token frequencies making cross-entropy mismatch a more useful proxy for model–data disagreement. Reconstruction scoring behaves differently because the residual is measured in the dequantised amplitude domain: uniform bins preserve a roughly linear amplitude scale, while CDF bins impose a non-linear map that can distort residual magnitudes.

The selected configuration ($K = 256$, token embed 32, `cdf` binning) provides the best likelihood PR-AUC while remaining competitive under reconstruction scoring, but overall gains remain modest (PR-AUC only slightly above prevalence). This suggests that the bottleneck is still score separability rather than only threshold tuning.

10.4 Windowing, smoothing, and the detection-latency trade-off

Two implementation details have first-order impact on real detection behaviour:

- **Window length and stride** determine which temporal structures are representable and how quickly scores update.
- **Causal smoothing** (trailing max) reduces nuisance spikes but adds detection delay that must be accounted for when interpreting latency.

This trade-off matters for detection quality, not just implementation convenience: high-impedance faults evolve slowly, so performance depends on balancing temporal context against latency and compute [1, 9].

Operational calibration outcomes. The calibrated-quantile operating point remains far from deployment-ready in this rerun:

- In the threshold-quantile sweep (Appendix A.1, Table A.1, Figure A.1), moving q from 0.99 to 0.997 changed calibrated thresholds but left operating outcomes nearly unchanged (F1 ≈ 0.409, FAR/day(normal) $\approx 2.5 \times 10^4$, detected fraction ≈ 0.062).
- In the alert-policy sweep (Appendix A.1, Table A.2, Figure A.2), increasing persistence from 1680 to 8400 samples reduced FAR/day by about $13.9\times$ ($\approx 2.5 \times 10^4 \rightarrow 1.8 \times 10^3$) but also reduced detected fraction from 0.062 to 0.021. Cooldown (0 vs 84000 samples) had negligible effect in this run.

These sweeps show that policy tuning mainly moves along a narrow trade-off frontier; it does not resolve the underlying score separability problem.

This is consistent with the distribution diagnostics (Chapter 8), where normal and fault score distributions overlap heavily and partially saturate, leaving no scalar threshold that cleanly separates operating regimes. The same pattern appears in the initial-noise ablation discussed in Section 10.2. The full-test follow-up run (`ablation_submission_best_checkpoint_full_eval_20260206_130720`) confirms this on 106,782 windows: reconstruction FAR/day(normal) is still $\approx 3.1 \times 10^3$ with 0/573 events detected, while likelihood detects only 23/573 events at FAR/day(normal) $\approx 7.3 \times 10^4$.

10.5 Evaluation limitations and threats to validity

Several limitations constrain how strongly the current results can be generalised:

- **Label uncertainty:** onset/offset times for faults are imperfect, especially for gradual faults; latency metrics therefore reflect both detector behaviour and label definitions.
- **Operating extrapolation under drift:** even with full-test evaluation, false-alarm/day rates can shift under long-term prevalence and drift changes not captured in a fixed offline split.

- **Distribution shift risk:** train/test directory splits reduce leakage but do not capture all deployment shifts (sensor hardware, mounting geometry, seasonal load changes) [14, 50].
- **Generative score ambiguity:** low ELBO-proxy loss (or low reconstruction error) does not guarantee in-distribution status, so score overlap can persist even when fidelity is high [34, 54, 58].

10.6 The role of contrastive learning

Contrastive learning with pseudo-anomalies shifts the objective from only modelling normality to explicitly separating normal from fault-like structure [46, 5, 16]. Introducing synthetic perturbations (for example, frequency injections or amplitude sags) during training can regularise the reconstruction manifold so that anomalous patterns are reconstructed less faithfully. Because pseudo-anomalies only approximate real grid faults, this approach should be treated as targeted regularisation rather than a substitute for real labelled faults.

10.7 Summary

Taken together, these experiments show that diffusion denoisers can produce interpretable diagnostic scores for structured, quasi-periodic time series in a one-class setting. The window-level distribution-learning framing remains reasonable, but the current discrete-diffusion scoring instantiation does not achieve fault–normal separability at realistic false-alarm budgets (full-test: reconstruction FAR/day(normal) $\approx 3.1 \times 10^3$ with $0/573$ events detected; likelihood FAR/day(normal) $\approx 7.3 \times 10^4$ with $23/573$ events detected).

Chapter 11

Conclusion and future work

This report evaluated diffusion-based generative models as *one-class* anomaly detectors for high-sample-rate magnetic-field time series in electrical distribution monitoring. The objective was to learn the distribution of normal, multi-modal, quasi-periodic behaviour from normal-only data and score fault-related deviations under drift, without relying on labelled faults for calibration.

11.1 Summary of contributions

The key contributions are:

- An end-to-end **discrete diffusion** pipeline for window-level distribution learning on multi-channel power signals, including quantisation (uniform and CDF/quantile binning) and a categorical denoiser trained on normal windows.
- A practical **window-scoring interface** suitable for streaming deployment, using (i) a single-pass reconstruction residual at a fixed corruption level and (ii) a Monte Carlo denoising-loss proxy for likelihood-style mismatch.
- A **deployment-realistic evaluation protocol** that prioritises normal-only threshold calibration and reports operating behaviour under a false-alarm budget, alongside threshold-free AUROC/PR-AUC.
- Implementations of additional scoring variants (contrastive pseudo-anomalies, gradient sensitivity, inpainting), establishing a base for follow-up work even where full evaluation is deferred.

11.2 Findings

The central empirical result is straightforward: diffusion models reconstruct normal structure well, but that does not consistently translate to strong detection. Across the completed model-selection sweep, the best configuration shows only modest gains over chance in sequence-level metrics (best likelihood-scoring AUROC 0.548, PR-AUC 0.336; prevalence 0.279), and reconstruction-based scoring is similarly limited (AUROC 0.515, PR-AUC 0.347). Quantisation interacts with score choice: CDF/quantile binning tends to help likelihood-style scoring (more balanced token frequencies), while uniform bins can help reconstruction PR-AUC when residuals are measured in the dequantised amplitude domain.

For deployment, the most important finding is that normal-only calibrated thresholds still produce unrealistically high false-alarm rates on the evaluated spans. In the full-test

follow-up run (`ablation_submission_best_checkpoint_full_eval_20260206_130720`), reconstruction scoring produced FAR/day(normal) $\approx 3.1 \times 10^3$ with 0/573 events detected, while likelihood-style scoring produced FAR/day(normal) $\approx 7.3 \times 10^4$ with 23/573 events detected. Persistence-based policies reduce false alarms but also suppress detections, which indicates that the main bottleneck is **score stability and separability** under drift, not a calibration-knob shortfall. This is consistent with the distribution diagnostics: normal and fault scores overlap heavily (with partial saturation), so no reasonable single threshold yields a deployment-acceptable operating point.

11.3 Future work

Because calibrated operating points remain impractical under heavy normal–fault overlap, the next experiments should first reshape score distributions, then optimise alerting policy.

1. **Geometry-aware corruption (top priority):** replace uniform random-replacement with locality-aware transition kernels for ordinal bins, and test whether early-fault separability improves in the deployment-relevant window/sequence views.
2. **Contrastive regularisation sweep (top priority):** run the planned pseudo-anomaly sweep over margin m and weight λ as targeted regularisation (not OOD "solution"), and report both separability diagnostics and calibrated FAR/day vs detected-event fraction.
3. **Two-stage score fusion (second priority):** keep reconstruction as a cheap first-stage score, then apply gradient/inpainting confirmatory checks on suspicious windows to reduce FAR/day at similar detection fraction.
4. **Windowing and smoothing study (second priority):** treat stride/window/smoothing settings as measurement trade-offs (latency vs alert load) after separability improves.
5. **Always-on diagnostics:** standardise normal-vs-fault distribution plots, fault-subtype splits, and score saturation checks for every run before interpreting threshold effects.

Biased timestep sampling remains a lower-priority empirical heuristic: it may help by allocating capacity to low-noise timesteps, but gains are not guaranteed and should be treated as ablation evidence rather than theory-backed certainty.

Overall, window-level distribution learning remains a sensible abstraction for this sensing modality, but the current discrete-diffusion scoring instantiation is not yet separable enough at realistic false-alarm budgets. Reconstruction fidelity alone is not enough for deployment readiness, so the next stage should prioritise anomaly-score separability and stability over additional model capacity.

Appendix A

Supplementary ablation results

This appendix collects supporting ablations referenced in Chapter 8. These results are useful for diagnostics and reproducibility but are not central to the main argument.

A.1 Operational sweeps from rerun-all (2026-02-05)

The rerun suite includes two practical sweeps targeting threshold and alert-policy sensitivity: `server_runs/ablation_threshold_calib_quantile_512bins_20260205_122315` and `server_runs/ablation_alert_policy_512bins_20260205_130449`.

Threshold calibration quantile sweep. Table A.1 and Figure A.1 show that changing the calibration quantile from $q = 0.99$ to $q = 0.997$ changes the calibrated reconstruction threshold ($\tau_{\text{rec}} = 0.521 \rightarrow 0.585$), but does *not* materially change the operating outcome in this setup: sequence F1 remains 0.409, FAR/day(normal) remains $\approx 2.50 \times 10^4$, and detected-event fraction remains 0.062. In this run, threshold quantile tuning alone was not the limiting factor. This aligns with the distribution diagnostics in Chapter 8: heavy normal–fault overlap and upper-range saturation imply that threshold movement alone cannot carve out a deployment-acceptable operating point; the failure mode is weak score separability, not quantile mis-calibration.

Table A.1: Threshold-calibration sweep (rerun-all): reconstruction sequence-level threshold and operating metrics.

q	τ_{rec}	F1 (seq)	FAR/day (normal)	Detected fraction
0.990	0.521	0.409	25021.6	0.062
0.995	0.551	0.409	25021.6	0.062
0.997	0.585	0.409	25021.6	0.062

Alert policy sweep. Table A.2 and Figure A.2 show that increasing `alert_persistence` from 1680 to 8400 samples reduces FAR/day(normal) from $\approx 2.50 \times 10^4$ to $\approx 1.80 \times 10^3$ (about 13.9$\times$ lower), but also reduces detected-event fraction from 0.062 to 0.021. Changing cooldown from 0 to 84000 samples has no visible effect in this run, indicating alerts are already sparse relative to the cooldown horizon. At these alert rates, further threshold-only tuning is unlikely to be acceptable in practice because analyst workload would still be dominated by false alarms [19, 48].

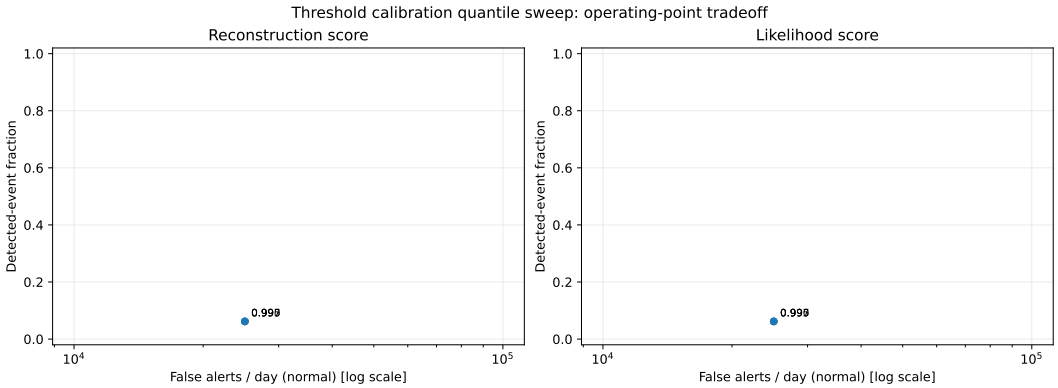


Figure A.1: Threshold-calibration sweep (rerun-all): FAR/day(normal) versus detected-event fraction for reconstruction and likelihood scores. Points are annotated by quantile q.

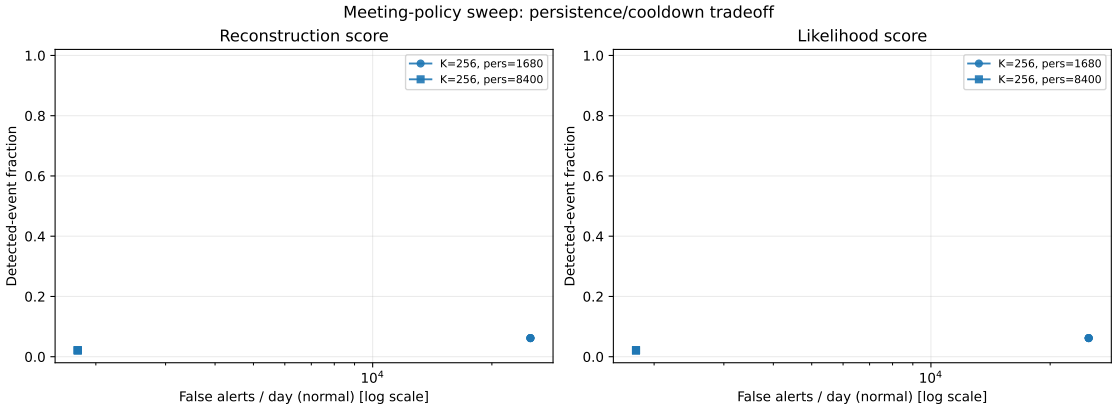


Figure A.2: Alert-policy sweep (rerun-all): FAR/day(normal) versus detected-event fraction. Lines connect cooldown settings within each $(K, \text{persistence})$ group.

Table A.2: Alert-policy sweep (rerun-all): reconstruction sequence-level operating metrics versus persistence/cooldown.

q	Persist (samples)	Cooldown (samples)	F1 (seq)	FAR/day (normal)	Detected fraction
0.997	1680	0	0.409	25021.6	0.062
0.997	1680	84000	0.409	25021.6	0.062
0.997	8400	0	0.409	1800.1	0.021
0.997	8400	84000	0.409	1800.1	0.021
0.999	1680	0	0.409	25021.6	0.062
0.999	1680	84000	0.409	25021.6	0.062
0.999	8400	0	0.409	1800.1	0.021
0.999	8400	84000	0.409	1800.1	0.021

Unavailable rerun sweep. The planned window-size-oracle rerun produced no valid metrics because configs still referenced an unresolved checkpoint path (`server_runs/ablation_window_sizes_oracle_eval_20260205_122220`).

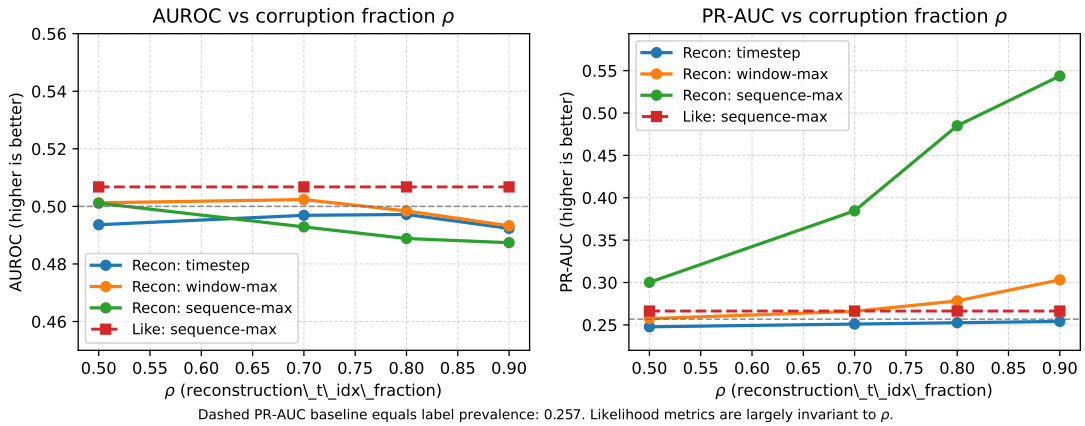


Figure A.3: Initial-noise sweep: threshold-free detection metrics vs reconstruction corruption fraction ρ. Reconstruction metrics change with ρ; likelihood metrics are largely invariant because they do not depend on the fixed t_{rec}. The dashed PR-AUC baseline equals the label prevalence.

A.2 Initial-noise sweep: is ρ=0.5 too low?

The reconstruction score is evaluated at a fixed corruption level $t_{\text{rec}} = \lfloor \rho T \rfloor$, where ρ corresponds to `evaluation.reconstruction_t_idx_fraction`. The motivating hypothesis is that $\rho = 0.5$ (the default setting used elsewhere in this report) may be *too low*: if the input is only moderately corrupted, a strong denoiser can reconstruct anomalous windows almost as well as normal windows, collapsing reconstruction-residual separation [54, 58].

The completed sweep in `server_runs/ablation_initial_noise_amount_20260205_112727` evaluates $\rho \in \{0.5, 0.7, 0.8, 0.9\}$ on a fixed checkpoint. Figure A.3 reports threshold-free AUROC/PR-AUC under reconstruction and likelihood scoring. Likelihood metrics are (as expected) largely invariant to ρ because they average denoising losses over random timesteps. Reconstruction PR-AUC increases monotonically with ρ (sequence-level PR-AUC 0.300 $\rightarrow$ 0.544), consistent with the hypothesis that $\rho = 0.5$ is too low for reconstruction-residual scoring in this setting. However, reconstruction AUROC remains near-random across the sweep (0.487–0.501), indicating that separability remains limited even when corruption is increased.

Figure A.4 explains the PR-AUC/AUROC mismatch: normal and fault distributions still overlap heavily and scores saturate near the upper residual bound. Figure A.5 and Table A.3 show the corresponding fidelity trade-off, while Figure A.6 provides fault-type splits.

Table A.3: Initial-noise sweep summary (fixed checkpoint): sequence-level reconstruction PR-AUC/AUROC and corresponding fidelity (MAE) for normal vs fault windows. τ_{cal} is the calibrated threshold under the normal-only quantile protocol.

ρ	t_{rec}	$\text{MAE}_{\text{norm}} \downarrow$	$\text{MAE}_{\text{fault}} \downarrow$	Recon PR-AUC$\uparrow$	Recon AUROC$\uparrow$	τ_{cal}
0.5	200	0.161	0.159	0.300	0.501	0.551
0.7	280	0.343	0.339	0.385	0.493	0.888
0.8	320	0.512	0.509	0.485	0.489	1.213
0.9	360	0.702	0.701	0.544	0.487	1.437

In this sweep, calibrated operating-point metrics are dominated by short-span extrapolation and score saturation effects. For this reason, threshold-free ranking metrics (AUROC/PR-

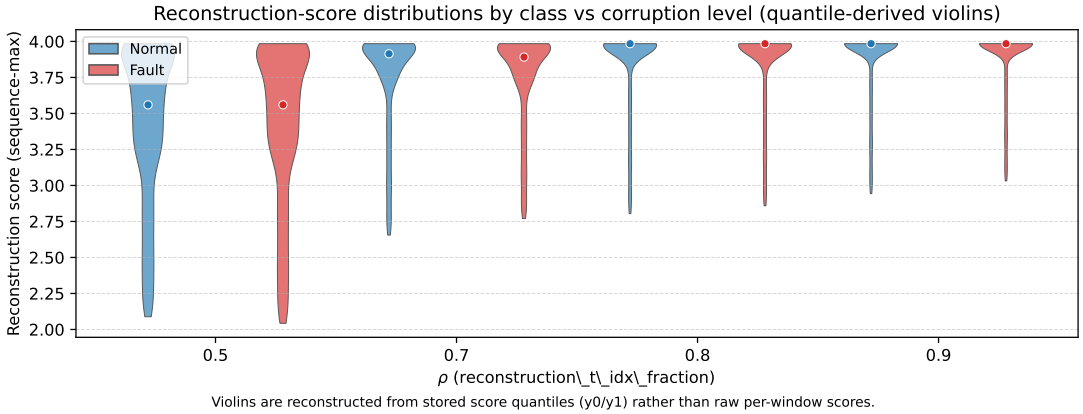


Figure A.4: Initial-noise sweep: reconstruction score distributions for normal vs fault windows (sequence-level sequence-max view). Violins are derived from stored score quantiles rather than raw per-window scores. Heavy overlap and upper-bound saturation help explain why separability remains limited even when PR-AUC increases with ρ.

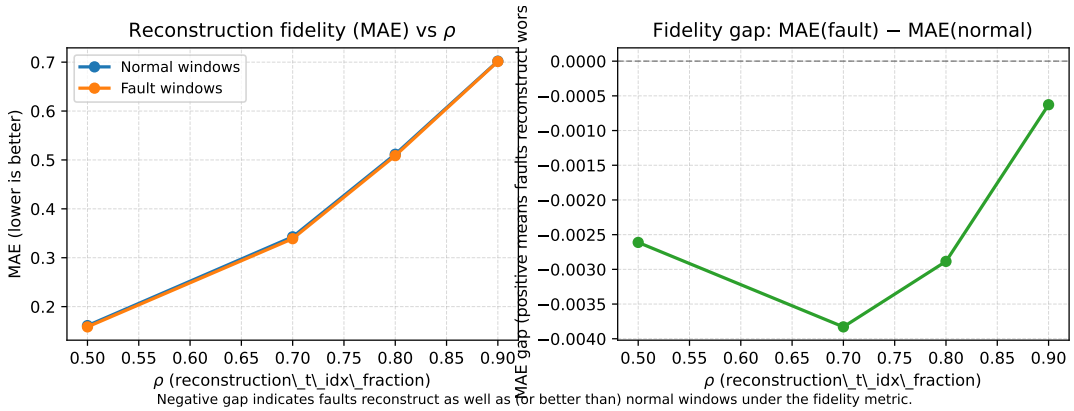


Figure A.5: Initial-noise sweep: reconstruction fidelity (MAE) vs ρ for normal and fault windows, and the resulting MAE gap. Negative gap means faults reconstruct as well as (or better than) normal windows under MAE.

AUC) are used as the primary evidence for the hypothesis test.

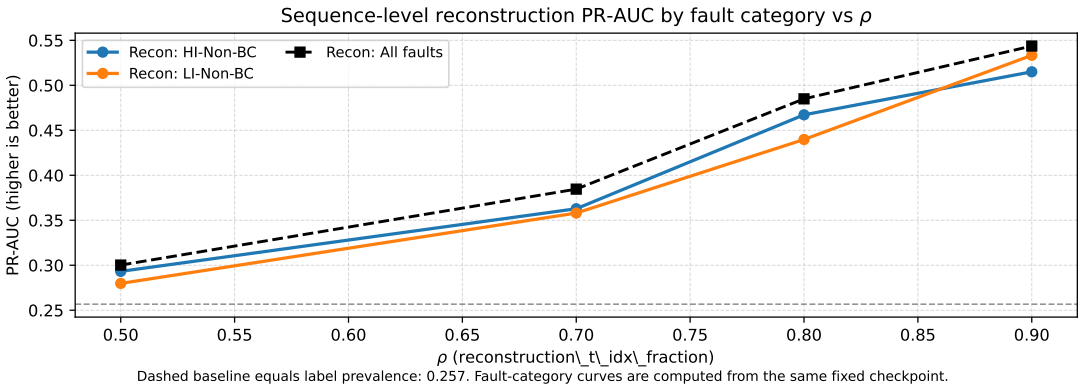


Figure A.6: Initial-noise sweep: reconstruction PR-AUC split by fault categories (where available in the dataset labels). The dashed baseline equals the overall label prevalence.

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